

Compute-Optimal Scaling Laws under Low-Rank Feature Map Perturbations

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Abstract

It has been observed that the generalization error (risk) of large language models on a variety of tasks often follows a power law trend with respect to model and dataset size. Moreover, the optimal choice of the latter hyperparameters with respect to a fixed compute budget also often obeys a power law. In this report, we address the attempts to theoretically explain this phenomenon under tractable assumptions. In particular, we focus on a paper that proves that these *compute-optimal scaling laws* arise in a student-teacher setting with a linear random features model and a linear target, where both the input covariance spectrum and signal's components satisfy power law decay with parameters α and β respectively. The authors show phase transitions between different regimes of power law scaling based on the value of (α, β) . We highlight the implications and limitations of this work, in addition to presenting some of its proofs in greater detail. Subsequently, we propose an extension where a rank-one perturbation containing the signal is added to the random feature map. We outline how proving scaling laws under this extension can leverage the results from this paper, and we empirically verify that faster power law decay occurs with this perturbation present.

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1 Introduction

In recent years, the focus of a large portion of the machine learning community has shifted to the study of large language models (LLMs). These systems, based on the Transformer neural network architecture [VSP⁺17], are trained on collections of large text corpora drawn from the Internet with the objective of predicting the next word in a text span. They can contain tens or even hundreds of billions of trainable parameters [BMR⁺20, SPN⁺22, AAA⁺23, ABA⁺23, TLI⁺23]. [BMR⁺20] in particular note that improved performance on downstream linguistic tasks has been consistently observed with increased model size (parameter count) in Transformer models.

Given the incredible computational expense of training ever-larger LLMs, the topic of *neural scaling laws* has been growing in popularity. The primary objective of work in this area is to predict the evolution of the generalization error (risk) on a given task as one scales the number of model parameters and training data observations [CJS⁺93, HNA⁺17, RRBS20, KMH⁺20, HBM⁺22, HKK⁺20, GDK21, GFF⁺22, HBC⁺22, ANZ22, CGRK22, SGS⁺22, MRB⁺23]. A natural byproduct is the question of *compute-optimality* [KMH⁺20, HBM⁺22]: as one scales a compute budget C , what proportion of it should be allocated towards increasing the number of parameters d as opposed to the training set size n ?

The study of risk trajectories with scaling of dataset size has been explored at least since [CJS⁺93], an empirical study focused on handwritten character recognition with the LeNet convolutional neural network on the NIST dataset [LBD⁺89]. The authors find that both the training and test error scale as a power law in n . More recently, [HNA⁺17] also observe power-law scaling of the test error with n for LSTMs [HS97] and ResNets [HZRS16] on language and vision tasks, respectively. Moreover, they find that the choice of d which achieves the best risk also obeys a power law in n . [RRBS20] follow up on this but shift the emphasis towards extrapolation (in other words, fitting a power law trend on smaller models and dataset sizes and using it to predict the risk for larger models).

The first scaling law work specifically on Transformer LLMs is [KMH⁺20]. To investigate the effect of d on the language modelling cross-entropy loss, LLMs of increasing size are trained to convergence. Similarly, the effect of n is characterized by fixing d sufficiently large so that it is not a bottleneck. The fitted power laws lead to an approximate characterization of the risk as

$$\mathcal{R}(d, n) \approx s \cdot n^{-\alpha_n} + \tilde{s} \cdot d^{-\alpha_d}. \quad (1)$$

On their custom enlargement of the WebText dataset [RWC⁺19], which consists of scraped text from outbound links on Reddit, the authors obtain $\alpha_n = 0.76$ and $\alpha_d = 0.077$. Moreover, defining compute by the formula $C = 6nd$, they find that the compute-optimal hyperparameter choice is $n \propto C^{0.73}$ and $d \propto C^{0.27}$. That is, according to their result, one should prioritize increasing language model size over dataset size.

[HBM⁺22] contradict the above, finding a better fit with $\alpha_n = \alpha_d = 0.5$ in their study of 400 LLMs with d ranging from 70 million to 16 billion and n ranging from 5 to 500 billion. Scaling n and d at the same rate has since been called *Chinchilla scaling*. The exact reason for their dramatically different result is not definitively identified but is speculated to be due to the combination of several factors: a different dataset (compilation of web pages, books, news articles, code on GitHub, and wikipedia articles), a different optimizer and learning rate scheduler, and a different tokenizer.

[KMH⁺20] and [HBM⁺22] inspired a plethora of empirical follow-up works investigating scaling

laws for downstream tasks. For example, [HKK⁺20] consider autoregressive generative modelling for language, vision, and mathematical problem solving. [GDK21, GFF⁺22] focus on machine translation, observing power-law scaling for both the cross-entropy loss and the popular BLEU score for assessing translation systems.

There is also a parallel line of work seeking to identify factors that alter the exponents in (or completely eliminate) power-law scaling. [HBC⁺22] find that power-law scaling is disrupted in the presence of repeated data (i.e., duplicate samples). Relatedly, [MRB⁺23] investigate compute-optimal scaling laws for LLMs trained on a dataset for multiple epochs. They identify three distinct phases of the risk trajectory: the first few epochs are essentially as effective as training with completely new data, then there is a period of diminishing returns until eventually there is no benefit of continuing for more epochs. [ANZ22] argue for a more flexible parametrization to fit the observed scaling behaviour rather than just a sum of power laws. [CGRK22] propose fitting with piecewise power laws (dubbed “broken” power laws) and obtain more accurate risk extrapolation on a suite of downstream tasks. [SGS⁺22] employ a data pruning strategy to select which examples to show the LLM during training and empirically show that this procedure leads to strictly better scaling than power law.

Of more direct interest to us are the attempts to build a theoretical understanding of neural scaling laws. The literature consists of works with varying levels of rigour in this regard. [SK20] non-rigorously argue that α_d should be of order $1/m^*$, where m^* is the dimension of an underlying manifold in which the data resides. They propose a heuristic to approximate this “intrinsic dimension” with a trained neural network applied to the data. [Hut21] considers a toy setting where inputs are drawn from a distribution over $\mathbb{N}$, and the student model predicts correctly if it has seen the input during training and incorrectly otherwise. If the data distribution follows a power law, i.e., the probability of each $i \in \mathbb{N}$ is proportional to $i^{-2\alpha}$ for some $\alpha > 1/2$, then the risk scales as a power law in the dataset size.

1.1 Power Law Random Features

The setup that we focus on in this report is one dubbed “power law random features” (PLRF) by [PPXP24]. We assume inputs of the form $\mathbf{x} \sim \mathcal{N}_m(\mathbf{0}, \mathbf{D})$, where $\mathbf{D} = \text{diag}(j^{-2\alpha} : j \in [m])$. This power law decay in the covariance spectrum of the input is frequently observed in real datasets [MRS22] and is often referred to as Zipf’s law in the context of natural language processing [Hut21]. Assume a linear teacher $f^* : \mathbb{R}^m \rightarrow \mathbb{R}$ such that $f^*(\mathbf{x}) = \langle \mathbf{b}, \mathbf{x} \rangle$ for some $\mathbf{b} \in \mathbb{R}^m$. We assume that the student model only has access to a random low-dimensional transformation of the input, $\mathbf{W}^T \mathbf{x}$, where $\mathbf{W} \in \mathbb{R}^{m \times d}$, $d < m$. [LWK⁺24] refer to $\mathbf{W}^T \mathbf{x}$ as “sketched covariates”. The student is a linear random features model which learns $\theta \in \mathbb{R}^d$ and outputs $f(\mathbf{x}) = \langle \theta, \mathbf{W}^T \mathbf{x} \rangle$. The risk is

$$\mathcal{R}_n(\theta, \mathbf{W}) := \mathbb{E}[(\langle \theta, \mathbf{W}^T \mathbf{x} \rangle - \langle \mathbf{b}, \mathbf{x} \rangle)^2 | \mathbf{W}] \quad (2)$$

Both [BDK⁺24]¹ and [MRS22] assume $\alpha > 1/2$ (so that the trace of $\mathbf{D}$ is finite) and $\mathbf{b} \sim \mathcal{N}_m(\mathbf{0}, \frac{1}{m} \mathbf{I}_m)$. They show that $\lim_{d \rightarrow \infty} \mathcal{R}_{n,d} \asymp n^{1-2\alpha}$ and $\lim_{n \rightarrow \infty} \mathcal{R}_{n,d} \asymp d^{1-2\alpha}$. [BDK⁺24] employ the replica method, a non-rigorous technique from statistical physics, whereas [MRS22] perform calculations involving random matrices with Feynman diagrams. Both of these works are agnostic to the training / model fitting procedure and work with the solution to OLS on the

¹Note that a preprint of this work first appeared in 2021 and in fact predates all other works in the PLRF setting that we discuss here.

training dataset.

Closer to the work that we focus on, [LWK⁺24] capture the case where $\mathbf{b}$ is also assumed to have power law covariance, i.e., $\mathbf{b} \sim \mathcal{N}_m(\mathbf{0}, \mathbf{H})$, where $\mathbf{H} = \text{diag}(j^{-2\beta} : j \in [m])$ for some $\beta > 0$. (Here, it is sufficient to have $2\alpha + 2\beta > 1$). They consider online stochastic gradient descent (i.e., a fresh independent minibatch of inputs is drawn at each iteration). Then, the scaling law of the risk lies in one of several regimes depending on the values of $(\alpha, \alpha + \beta)$. Their analysis is founded on a rigorous decomposition of the excess risk into approximation, bias, and variance errors.

Finally, [BAP24a] perform an analysis in a very similar setting to the above and also identify regimes based on α, β . Their analysis relies on the physics-inspired dynamical mean field theory (DMFT). Their recent follow-up work [BAP24a] extends this to a more flexible student that can perform limited “feature learning”. That is, it can learn a map $\mathbf{B} \in \mathbb{R}^{d \times d}$ such that $f(\mathbf{x}) = \langle \boldsymbol{\theta}, \mathbf{B} \mathbf{W}^T \mathbf{x} \rangle$. Once again using DMFT (and also in empirical simulations), they show that if the regression task is sufficiently “hard” (in our notation, if β is sufficiently small), then the scaling law exponent improves over the random features case.

This report is focused on the recent work of [PPXP24]², who work in a very similar setting to the papers above. Specifically, they keep the same assumptions on the input but assume that $\mathbf{b}$ is deterministic such that $b_j = j^{-\beta}$ for each $j \in [m]$. They demonstrate four “phases” (and three “subphases”) of risk scaling based on α and β . Their analysis is entirely based on random matrix theory. In this report, we provide an overview of the setting and the key assumptions made, an outline of the analysis, illustrative proofs of some of the key results, and replicate some of the empirical simulations. Moreover, we propose an extension with the aim of pushing this line of work towards a rigorous treatment of feature learning, which is widely believed to be a crucial ingredient in the success of neural network models [BCV13, YH21].

Notation. $\langle \cdot, \cdot \rangle$ denotes the Euclidean inner product for vectors and the Frobenius inner product for matrices, and $[n]$ denotes the set $\{1, \dots, n\}$. In the absence of a subscript, $\|\cdot\|$ denotes the Euclidean norm for vectors and the operator norm for matrices. For a vector $\mathbf{v}$, we write $\mathbf{v}^{\otimes 2} := \mathbf{v} \mathbf{v}^T$. For a square matrix $\mathbf{M}$, we denote its spectrum by $\text{spec}(\mathbf{M})$. The notation $\xrightarrow{\mathbb{P}}$ denotes convergence in probability. Following [PPXP24], we write $f(n, d) \lesssim g(n, d)$ if there exist $C > 0, n_0, d_0$ such that for all $n \geq n_0, d \geq d_0$ we have $f(n, d) \leq C \cdot g(n, d)$. We define the notation $f(n, d) \gtrsim g(n, d)$ similarly. If the above two properties hold, then we write $f(n, d) \asymp g(n, d)$.

2 A Random Matrix Analysis of Scaling Laws

We begin by re-stating the PLRF setting introduced in the previous section. We define a *data complexity* $\alpha > 0$ and *target complexity* $\beta > 0$. We assume that $2\alpha + 2\beta > 1$ throughout this report. As we will see, α and β control the difficulty of the regression problem. We assume input $\mathbf{x} \sim \mathcal{N}_m(\mathbf{0}, \mathbf{D})$, where $\mathbf{D} := \text{diag}(j^{-2\alpha} : j \in [m])$. We note that, of the related papers mentioned in the previous section, [BDK⁺24, BAP24a, BAP24b] do not assume normality. We assume the target is noiseless³ and takes the form $y = \langle \mathbf{b}, \mathbf{x} \rangle$, where $\mathbf{b} = (b_1, \dots, b_m) \in \mathbb{R}^m$. Moreover, $\mathbf{b}$ is taken deterministic with power law decay $b_j = j^{-\beta}, j \in [m]$.

We distort the straightforward regression setup above by additionally assuming that the student

²At the time of writing, this paper is only available as a preprint but will appear at the upcoming NeurIPS 2024 conference.

³[PPXP24] remark that additive noise does not change the scaling laws unless d and r are sufficiently large so that the irreducible error supersedes other sources of error. Once this point is reached, there is no longer power law scaling and the learning curve becomes constant.

model has access only to the d -dimensional features $\mathbf{W}^T \mathbf{x}$, where $\mathbf{W} \in \mathbb{R}^{m \times d}$ has i.i.d. $\mathcal{N}(0, \frac{1}{d})$ entries. We work in the asymptotic regime where $m, d \rightarrow \infty$ such that $m/d \rightarrow c > 1$. The student is then taken to be a linear model on top of the sketched features: $\hat{y} = \langle \boldsymbol{\theta}, \mathbf{W}^T \mathbf{x} \rangle$ for a parameter vector $\boldsymbol{\theta} \in \mathbb{R}^d$. The latter is initialized at zero and updated via online SGD for r steps with step size γ and batch size B :

$$\boldsymbol{\theta}_{r+1} = \boldsymbol{\theta}_r - \gamma \sum_{i=1}^B \mathbf{W}^T \mathbf{x}_r^{(i)} (\langle \mathbf{W}^T \mathbf{x}_r^{(i)}, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}_r^{(i)}, \mathbf{b} \rangle), \quad (3)$$

where $\mathbf{x}_r^{(1)}, \dots, \mathbf{x}_r^{(B)}$ are i.i.d. draws from the input distribution. Assume that $\gamma B < 1$ for all m, d .

To characterize the computational cost of training the student, we follow the convention set by [KMH⁺20], in defining the training compute (flops) as $C = rBd$.⁴ We are interested in finding the optimal choice of d (and thus of r) given a fixed compute budget C and batch size B to minimize the risk

$$\mathcal{R}_d(\boldsymbol{\theta}; \mathbf{W}) := \mathbb{E}[(\langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta} \rangle - \langle \mathbf{x}, \mathbf{b} \rangle)^2 | \mathbf{W}]. \quad (4)$$

The compute-optimal choice of d is

$$d^*(C) \in \arg \min_d \{ \mathcal{R}_d(\boldsymbol{\theta}_r; \mathbf{W}) : C = rBd \}. \quad (5)$$

The aforementioned empirical studies of scaling laws [KMH⁺20, HBM⁺22] find that the relationship between compute and the number of parameters in large Transformer language models can be reliably fit with a power law, i.e., $d^* = \tilde{s}C^{-\xi}$. With this compute-optimal choice, the risk can then also be fit with a power law, i.e., $\mathcal{R}_{d^*} = sC^{-\eta}$. [PPXP24] show that this is the case under their theoretical model as well.

2.1 SGD Dynamics

The first key ingredient to the analysis is a characterization of the dynamics of $\mathcal{R}_d$ under SGD.

Proposition 2.1 (Volterra Equation for SGD Dynamics, [PPXP24]). *Let $\hat{\mathbf{K}}_0 = \mathbf{D}^{1/2} \mathbf{W} \mathbf{W}^T \mathbf{D}^{1/2}$ and $\mathbf{a} = \mathbf{D}^{1/2} \mathbf{b}$. Then, the risk incurred by the student model after r iterations of online stochastic gradient descent satisfies the following convolution-type Volterra integral equation:*

$$\mathcal{R}_d(\boldsymbol{\theta}_r) = \mathcal{F}(r) + \mathcal{K} * \mathcal{R}_d(\boldsymbol{\theta}_r), \quad (6)$$

where the forcing function $\mathcal{F}$ and kernel function $\mathcal{K}$ are defined on the nonnegative integers by

$$\mathcal{F}(r) := -\frac{1}{2\pi i} \oint_{\Gamma} \mathbf{a}^T (\hat{\mathbf{K}}_0 - z)^{-1} \mathbf{a} (1 - 2\gamma Bz + 2\gamma^2 B^2 z^2)^r dz, \quad (7)$$

$$\mathcal{K}(r) := -\frac{1}{2\pi i} \text{tr} \left(\oint_{\Gamma} (\hat{\mathbf{K}}_0 - z)^{-1} z^2 (1 - 2\gamma Bz + 2\gamma^2 B^2 z^2)^r dz \right) \quad (8)$$

for any contour Γ enclosing the spectrum of $\hat{\mathbf{K}}_0$, and the convolution operation $*$ is defined by $(\mathcal{K} * f)(r) = \sum_{s=0}^{r-1} \mathcal{K}(r-1-s)f(s)$ for any f defined on $\mathbb{N}_0$.

⁴[KMH⁺20] take $C = 6rBd$, we omit the constant here. Note also that since we are doing online SGD, the dataset size n is simply Br .

Proof. We cover the entire proof from [PPXP24] in detail. We re-organize some steps for easier readability and fill in gaps where steps were skipped.

We begin by expanding the risk as

$$\begin{aligned}
\mathcal{R}(\boldsymbol{\theta}_r) &= \mathbb{E}[(\langle \boldsymbol{\theta}_r, \mathbf{W}^T \mathbf{x} \rangle - \langle \mathbf{b}, \mathbf{x} \rangle)^2 | \mathbf{W}] \\
&= \mathbb{E}[\boldsymbol{\theta}_r^T \mathbf{W}^T \mathbf{x} \mathbf{x}^T \boldsymbol{\theta}_r - 2\boldsymbol{\theta}_r^T \mathbf{W}^T \mathbf{x} \mathbf{x}^T \mathbf{b} + \mathbf{b}^T \mathbf{x} \mathbf{x}^T \mathbf{b} | \mathbf{W}] \\
&= \boldsymbol{\theta}_r^T \mathbf{W}^T \mathbf{D} \mathbf{W} \boldsymbol{\theta}_r - 2\boldsymbol{\theta}_r^T \mathbf{W}^T \mathbf{D} \mathbf{b} + \mathbf{b}^T \mathbf{D} \mathbf{b} \\
&= \|\mathbf{D}^{1/2}(\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})\|_2^2.
\end{aligned} \tag{9}$$

Note that $\mathbb{R}^m = \text{Im}(\mathbf{W}) \oplus \text{Ker}(\mathbf{W}^T \mathbf{D})$. Indeed, consider the inner product defined by $\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{D}} := \mathbf{u}^T \mathbf{D} \mathbf{v}$. Let $\mathbf{u} \in \text{col}(\mathbf{W})$, $\mathbf{v} \in \text{Ker}(\mathbf{W}^T \mathbf{D})$. Then $\mathbf{u} = \mathbf{W} \tilde{\mathbf{u}}$ for some $\tilde{\mathbf{u}} \in \mathbb{R}^d$, and $\mathbf{W}^T \mathbf{D} \mathbf{v} = \mathbf{0}$. Then,

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{D}} = \mathbf{u}^T \mathbf{D} \mathbf{v} = (\mathbf{W} \tilde{\mathbf{u}})^T \mathbf{D} \mathbf{v} = \tilde{\mathbf{u}}^T \mathbf{W}^T \mathbf{D} \mathbf{v} = \mathbf{v} = 0. \tag{10}$$

Hence, we may write

$$\mathbf{b} = \mathbf{W} \check{\mathbf{b}} + \dot{\mathbf{b}} \tag{11}$$

for some $\check{\mathbf{b}} \in \mathbb{R}^d$ and $\dot{\mathbf{b}} \in \mathbb{R}^m$ satisfying $\mathbf{W}^T \mathbf{D} \dot{\mathbf{b}} = \mathbf{0}$. Letting $\{\lambda_j\}_{j=1}^m$ be the eigenvalues of $\mathbf{W}^T \mathbf{D} \mathbf{W}$ (in nondecreasing order) and $\{\mathbf{w}_j\}_{j=1}^m$ be associated normalized eigenvectors, we have

$$\begin{aligned}
\mathcal{R}(\boldsymbol{\theta}_r) &= \boldsymbol{\theta}_r^T (\mathbf{W}^T \mathbf{D} \mathbf{W}) \boldsymbol{\theta}_r - 2\boldsymbol{\theta}_r^T \mathbf{W}^T \mathbf{D} \mathbf{W} \check{\mathbf{b}} + (\mathbf{W} \check{\mathbf{b}} + \dot{\mathbf{b}})^T \mathbf{D} (\mathbf{W} \check{\mathbf{b}} + \dot{\mathbf{b}}) \\
&= \boldsymbol{\theta}_r^T \left(\sum_{j=1}^m \lambda_j \mathbf{w}_j \mathbf{w}_j^T \right) \boldsymbol{\theta}_r - 2\boldsymbol{\theta}_r^T \left(\sum_{j=1}^m \lambda_j \mathbf{w}_j \mathbf{w}_j^T \right) \check{\mathbf{b}} + \check{\mathbf{b}}^T \left(\sum_{j=1}^m \lambda_j \mathbf{w}_j \mathbf{w}_j^T \right) \check{\mathbf{b}} + \dot{\mathbf{b}}^T \mathbf{D} \dot{\mathbf{b}} \\
&= \sum_{j=1}^m \lambda_j \rho_j^2 + \|\mathbf{D}^{1/2} \dot{\mathbf{b}}\|_2^2,
\end{aligned} \tag{12}$$

where we define

$$\rho_j^2(r) := (\boldsymbol{\theta}_r - \check{\mathbf{b}})^T \mathbf{w}_j \mathbf{w}_j^T (\boldsymbol{\theta}_r - \check{\mathbf{b}}) = \langle \mathbf{w}_j \mathbf{w}_j, (\boldsymbol{\theta}_r - \check{\mathbf{b}})^{\otimes 2} \rangle. \tag{13}$$

Hence, the analysis reduces to the dynamics of the $\rho_j^2(r)$. In particular, we wish to understand

$$\mathbb{E}[\rho_j^2(r+1) - \rho_j^2(r) | \mathcal{F}_r], \tag{14}$$

where $\mathcal{F}_r = \sigma(\mathbf{W}, \boldsymbol{\theta}_0, \dots, \boldsymbol{\theta}_r)$, i.e., the σ -algebra generated by the random feature map and the SGD iterates up to step r . We have the following result, whose proof we defer to Appendix A.

Lemma 2.2 (Quadratic Update). *Let $q : \mathbb{R}^d \rightarrow \mathbb{R}$ be a quadratic function. Then,*

$$\begin{aligned}
\mathbb{E}[q(\boldsymbol{\theta}_{r+1}) - q(\boldsymbol{\theta}_r) | \mathcal{F}_r] &= -\gamma B \langle \nabla q(\boldsymbol{\theta}_r), \mathbf{W}^T \mathbf{D} \mathbf{W} (\boldsymbol{\theta} - \check{\mathbf{b}}) \rangle \\
&\quad + \frac{\gamma^2 B}{2} \langle \nabla^2 q, \mathbf{W}^T \mathbf{D} \mathbf{W} \rangle \|\mathbf{D}^{1/2}(\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})\|_2^2 \\
&\quad + \gamma^2 B^2 \langle \mathbf{W}^T \mathbf{D} \mathbf{W} (\nabla^2 q) \mathbf{W}^T \mathbf{D} \mathbf{W}, (\boldsymbol{\theta}_r - \check{\mathbf{b}})^{\otimes 2} \rangle.
\end{aligned} \tag{15}$$

In our case,

$$\rho_j^2(r) = (\boldsymbol{\theta}_r - \check{\mathbf{b}})^T \mathbf{w}_j \mathbf{w}_j^T (\boldsymbol{\theta}_r - \check{\mathbf{b}}), \quad \nabla_{\boldsymbol{\theta}} \rho_j^2(r) = 2\mathbf{w}_j \langle \mathbf{w}_j, \boldsymbol{\theta}_r - \check{\mathbf{b}} \rangle, \quad \nabla^2 \rho_j^2(r) = 2\mathbf{w}_j \mathbf{w}_j^T. \tag{16}$$

Plugging $q(\boldsymbol{\theta}_r) = \rho_j^2(r)$ into (97), we obtain

$$\begin{aligned}\mathbb{E}[\rho_j^2(r+1) - \rho_j^2(r)|\mathcal{F}_r] &= -2\gamma B \langle \mathbf{w}_j, \boldsymbol{\theta}_r - \check{\mathbf{b}} \rangle \langle \mathbf{w}_j, \mathbf{W}^T \mathbf{D} \mathbf{W} (\boldsymbol{\theta} - \check{\mathbf{b}}) \rangle \\ &\quad + B\gamma^2 \langle \mathbf{w}_j \mathbf{w}_j^T, \mathbf{W}^T \mathbf{D} \mathbf{W} \rangle \|\mathbf{D}^{1/2}(\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})\|_2^2 \\ &\quad + 2B^2\gamma^2 \langle \mathbf{W}^T \mathbf{D} \mathbf{W} \mathbf{w}_j \mathbf{w}_j^T \mathbf{W}^T \mathbf{D} \mathbf{W}, (\boldsymbol{\theta}_r - \check{\mathbf{b}})^{\otimes 2} \rangle \\ &= -2\gamma B \lambda_j \rho_j^2(r) + \gamma^2 B \lambda_j \|\mathbf{D}^{1/2}(\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})\|_2^2 + 2\gamma^2 B^2 \lambda_j^2 \rho_j^2(r),\end{aligned}\tag{17}$$

where the last equality uses the fact that $\mathbf{w}_j$ is a unit-norm eigenvector of $\mathbf{W}^T \mathbf{D} \mathbf{W}$ associated with the eigenvalue λ_j . Moreover, by the tower property,

$$\mathbb{E}[\rho_j^2(r+1)|\mathbf{W}] = (1 - 2\gamma B \lambda_j + 2\gamma^2 B^2 \lambda_j^2) \mathbb{E}[\rho_j^2(r)|\mathbf{W}] + \gamma^2 B \lambda_j \mathcal{R}(\boldsymbol{\theta}_r).\tag{18}$$

It is straightforward to show by induction that the recurrence is solved by

$$\begin{aligned}\rho_j^2(r) &= (1 - 2\gamma B \lambda_j + 2\gamma^2 B^2 \lambda_j^2)^r \rho_j^2(0) \\ &\quad + \gamma^2 B \sum_{s=0}^{r-1} (1 - 2\gamma B \lambda_j + 2\gamma^2 B^2 \lambda_j^2)^{r-1-s} \lambda_j \mathcal{R}(\boldsymbol{\theta}_s).\end{aligned}\tag{19}$$

Then, plugging this into (12), we have

$$\begin{aligned}\mathcal{R}(\boldsymbol{\theta}_r) &= \sum_{j=1}^d \lambda_j \rho_j^2(0) (1 - 2\gamma B \lambda_j + 2\gamma^2 B^2 \lambda_j^2)^r + \|\mathbf{D}^{1/2} \dot{\mathbf{b}}\|_2^2 \\ &\quad + \sum_{j=1}^d \gamma^2 B \lambda_j^2 \sum_{s=0}^{r-1} (1 - 2\gamma B \lambda_j + 2\gamma^2 B^2 \lambda_j^2)^{r-1-s} \mathcal{R}(\boldsymbol{\theta}_s).\end{aligned}\tag{20}$$

Letting $\check{\mathbf{K}}_0 := \mathbf{W}^T \mathbf{D} \mathbf{W}$, we may simplify the first term of the RHS above as

$$\begin{aligned}\sum_{j=1}^d \lambda_j \rho_j^2(0) (1 - 2\gamma B \lambda_j + 2\gamma^2 B^2 \lambda_j^2)^r &= \sum_{j=1}^d \langle \lambda_j \mathbf{w}_j \mathbf{w}_j^T (1 - 2\gamma B \lambda_j + 2\gamma^2 B^2 \lambda_j^2)^r, (\boldsymbol{\theta}_0 - \check{\mathbf{b}})^{\otimes 2} \rangle \\ &= \langle \check{\mathbf{K}}_0 (\mathbf{I} - 2\gamma B \check{\mathbf{K}}_0 + 2\gamma^2 B^2 \check{\mathbf{K}}_0^2)^r, (\boldsymbol{\theta}_0 - \check{\mathbf{b}})^{\otimes 2} \rangle.\end{aligned}\tag{21}$$

Since it is easier (from the random matrix theory perspective) to work with covariance matrices, we convert this expression to one depending on $\hat{\mathbf{K}}_0 := \mathbf{D}^{1/2} \mathbf{W} \mathbf{W}^T \mathbf{D}^{1/2}$. Consider the singular value decomposition

$$\mathbf{D}^{1/2} \mathbf{W} = \mathbf{V} \boldsymbol{\Sigma} \mathbf{U}^T, \quad \check{\mathbf{K}}_0 = \mathbf{U} \boldsymbol{\Sigma}^T \boldsymbol{\Sigma} \mathbf{U}^T, \quad \hat{\mathbf{K}}_0 = \mathbf{V} \boldsymbol{\Sigma} \boldsymbol{\Sigma}^T \mathbf{V}^T.\tag{22}$$

Then,

$$\begin{aligned}\langle \check{\mathbf{K}}_0 \text{poly}(\check{\mathbf{K}}_0), (\boldsymbol{\theta}_0 - \check{\mathbf{b}})^{\otimes 2} \rangle &= \langle \mathbf{U} \boldsymbol{\Sigma}^T \boldsymbol{\Sigma} \text{poly}(\boldsymbol{\Sigma}^T \boldsymbol{\Sigma}) \mathbf{U}^T, (\boldsymbol{\theta}_0 - \check{\mathbf{b}})^{\otimes 2} \rangle \\ &= \langle \text{poly}(\boldsymbol{\Sigma}^T \boldsymbol{\Sigma}) \boldsymbol{\Sigma}^T \boldsymbol{\Sigma}, (\mathbf{U}^T (\boldsymbol{\theta}_0 - \check{\mathbf{b}}))^{\otimes 2} \rangle \\ &= \text{tr}(\boldsymbol{\Sigma}^T \boldsymbol{\Sigma} \text{poly}(\boldsymbol{\Sigma}^T \boldsymbol{\Sigma}) (\mathbf{U}^T (\boldsymbol{\theta}_0 - \check{\mathbf{b}}))^{\otimes 2}) \\ &= \text{tr}(\mathbf{V} \boldsymbol{\Sigma} \text{poly}(\boldsymbol{\Sigma}^T \boldsymbol{\Sigma}) (\mathbf{U} (\boldsymbol{\theta}_0 - \check{\mathbf{b}}))^{\otimes 2} \boldsymbol{\Sigma}^T \mathbf{V}^T) \\ &= \text{tr}(\mathbf{V} \text{poly}(\boldsymbol{\Sigma} \boldsymbol{\Sigma}^T) \boldsymbol{\Sigma} (\mathbf{U} (\boldsymbol{\theta}_0 - \check{\mathbf{b}}))^{\otimes 2} \boldsymbol{\Sigma}^T \mathbf{V}^T)\end{aligned}$$

$$\begin{aligned}
&= \text{tr}(\mathbf{V} \text{poly}(\mathbf{\Sigma} \mathbf{\Sigma}^T) \mathbf{V}^T (\mathbf{V} \mathbf{\Sigma} \mathbf{U}^T (\boldsymbol{\theta}_0 - \check{\mathbf{b}}))^{\otimes 2}) \\
&= \langle \mathbf{V} \text{poly}(\mathbf{\Sigma} \mathbf{\Sigma}^T) \mathbf{V}^T, (\mathbf{V} \mathbf{\Sigma} \mathbf{U}^T (\boldsymbol{\theta}_0 - \check{\mathbf{b}}))^{\otimes 2} \rangle \\
&= \langle \text{poly}(\hat{\mathbf{K}}_0), (\mathbf{D}^{1/2} \mathbf{W} (\boldsymbol{\theta}_0 - \check{\mathbf{b}}))^{\otimes 2} \rangle.
\end{aligned}$$

Plugging back into (21), we have

$$\sum_{j=1}^d \lambda_j \rho_j^2(0) (1 - 2\gamma B \lambda_j + 2\gamma^2 B^2 \lambda_j^2)^r = \langle (\mathbf{I} - 2\gamma B \hat{\mathbf{K}}_0 + 2\gamma^2 B^2 \hat{\mathbf{K}}_0^2)^r, (\mathbf{D}^{1/2} \mathbf{W} (\boldsymbol{\theta}_0 - \check{\mathbf{b}}))^{\otimes 2} \rangle. \quad (23)$$

Moreover, we have

$$\sum_{j=1}^d \gamma^2 B \lambda_j^2 (1 - 2\gamma B \lambda_j + 2\gamma^2 B^2 \lambda_j^2)^{r-1-s} = \gamma^2 B \text{tr}(\hat{\mathbf{K}}_0^2 (\mathbf{I} - 2\gamma B \hat{\mathbf{K}}_0 + 2\gamma^2 B^2 \hat{\mathbf{K}}_0^2)^{r-1-s}), \quad (24)$$

and we denote this by $\mathcal{K}(r-1-s)$. Hence, combining (20), (23), and (24), we arrive at the convolution-type Volterra equation

$$\begin{aligned}
\mathcal{R}(\boldsymbol{\theta}_r) &= \langle (\mathbf{I} - 2\gamma B \hat{\mathbf{K}}_0 + 2\gamma^2 B^2 \hat{\mathbf{K}}_0^2)^r, (\mathbf{D}^{1/2} \mathbf{W} (\boldsymbol{\theta}_0 - \check{\mathbf{b}}))^{\otimes 2} \rangle + \|\mathbf{D}^{1/2} \dot{\mathbf{b}}\|_2^2 \\
&\quad + \sum_{s=0}^{r-1} \mathcal{K}(r-1-s) \cdot \mathcal{R}(\boldsymbol{\theta}_s).
\end{aligned} \quad (25)$$

$$\begin{aligned}
\mathcal{F}(r) &:= \langle (\mathbf{I} - 2\gamma B \hat{\mathbf{K}}_0 + 2\gamma^2 B^2 \hat{\mathbf{K}}_0^2)^r, (\mathbf{D}^{1/2} \mathbf{W} (\boldsymbol{\theta}_0 - \check{\mathbf{b}}))^{\otimes 2} \rangle + \|\mathbf{D}^{1/2} \dot{\mathbf{b}}\|_2^2 \\
&= \check{\mathbf{b}}^T \mathbf{W}^T \mathbf{D}^{1/2} (\mathbf{I} - 2\gamma B \hat{\mathbf{K}}_0 + 2\gamma^2 B^2 \hat{\mathbf{K}}_0^2)^r \mathbf{D}^{1/2} \mathbf{W} \check{\mathbf{b}} + \dot{\mathbf{b}}^T \mathbf{D} \dot{\mathbf{b}} \\
&= (\mathbf{b} - \dot{\mathbf{b}})^T \mathbf{D}^{1/2} \mathbf{V} (\mathbf{I} - 2\gamma B \mathbf{\Sigma} \mathbf{\Sigma}^T + 2\gamma^2 B^2 (\mathbf{\Sigma} \mathbf{\Sigma}^T)^2)^r \mathbf{V}^T \mathbf{D}^{1/2} (\mathbf{b} - \dot{\mathbf{b}}) + \dot{\mathbf{b}}^T \mathbf{D} \dot{\mathbf{b}}.
\end{aligned} \quad (26)$$

Since $\mathbf{W}^T \mathbf{D} \dot{\mathbf{b}} = 0$, we have

$$\mathbf{U} \mathbf{\Sigma}^T \mathbf{V}^T \mathbf{D}^{1/2} \dot{\mathbf{b}} = \mathbf{0} \Rightarrow \mathbf{\Sigma}^T \mathbf{V}^T \mathbf{D}^{1/2} \dot{\mathbf{b}} = \mathbf{0}, \quad (27)$$

and therefore plugging back into (26), we have

$$\mathcal{F}(r) = \mathbf{b}^T \mathbf{D}^{1/2} (\mathbf{I} - 2\gamma B \hat{\mathbf{K}}_0 + 2\gamma^2 B^2 \hat{\mathbf{K}}_0^2)^r \mathbf{D}^{1/2} \mathbf{b} + 2\dot{\mathbf{b}}^T \mathbf{D} \dot{\mathbf{b}} - \dot{\mathbf{b}}^T \mathbf{D} \mathbf{b} - \mathbf{b}^T \mathbf{D} \dot{\mathbf{b}}. \quad (28)$$

Now, noticing that

$$\mathbf{b}^T \mathbf{D} \dot{\mathbf{b}} = (\mathbf{W} \check{\mathbf{b}} + \dot{\mathbf{b}})^T \mathbf{D} \dot{\mathbf{b}} = \check{\mathbf{b}}^T \mathbf{W}^T \mathbf{D} \dot{\mathbf{b}} + \dot{\mathbf{b}}^T \mathbf{D} \dot{\mathbf{b}} = \dot{\mathbf{b}}^T \mathbf{D} \dot{\mathbf{b}}, \quad (29)$$

we have

$$\mathcal{F}(r) = \mathbf{b}^T \mathbf{D}^{1/2} (\mathbf{I} - 2\gamma B \hat{\mathbf{K}}_0 + 2\gamma^2 B^2 \hat{\mathbf{K}}_0^2)^r \mathbf{D}^{1/2} \mathbf{b}. \quad (30)$$

To allow for the use of tools from random matrix theory, we reformulate this in terms of the resolvent of $\hat{\mathbf{K}}_0$, i.e., the matrix $(\hat{\mathbf{K}}_0 - z)^{-1}$ for each $z \in \mathbb{C} \setminus \text{spec}(\hat{\mathbf{K}}_0)$. We further initialize $\boldsymbol{\theta}_0 = \mathbf{0}$. By an application of Cauchy's integral formula in the context of matrix resolvents (e.g., see Section

2.1.3 of [CL22]), we have

$$\mathcal{F}(r) = \oint_{\Gamma} \mathbf{b}^T \mathbf{D}^{1/2} (\hat{\mathbf{K}}_0 - z)^{-1} \mathbf{D}^{1/2} \mathbf{b}^T (1 - 2\gamma Bz + 2\gamma B^2 z^2)^r dz, \quad (31)$$

where Γ is any contour enclosing the spectrum of $\hat{\mathbf{K}}_0$. Similarly, we obtain

$$\begin{aligned} \mathcal{K}(r) &= \gamma^2 B \operatorname{tr} (\hat{\mathbf{K}}_0^2 (\mathbf{I} - 2\gamma B \hat{\mathbf{K}}_0 + 2\gamma^2 B^2 \hat{\mathbf{K}}_0^2)^r) \\ &= -\frac{1}{2\pi i} \gamma^2 B \oint_{\Gamma} z^2 (1 - 2\gamma Bz + 2\gamma^2 B^2 z^2)^r \operatorname{tr} (\hat{\mathbf{K}}_0 - z)^{-1} dz. \end{aligned} \quad (32)$$

□

Remark 2.3. [PPXP24] interpret the kernel function as capturing the contribution of SGD noise to the excess risk. To see this, notice that if we consider full-batch gradient descent with dataset size n scaling with d and r , then the assumption $\gamma B < 1$ implies $\gamma^2 B \lesssim n^{-1}$. However, in our SGD setting, we assume γ and B are of constant order.

The randomness of $\mathbf{W}$ creates difficulties in analyzing $\mathcal{F}$ and $\mathcal{K}$. [PPXP24] consider an approximating deterministic Volterra equation by replacing the random resolvent $(\hat{\mathbf{K}}_0 - z)^{-1}$ appearing in $\mathcal{F}$ and $\mathcal{K}$ by its *deterministic equivalent*. The latter refers to a matrix $\mathbf{R}_0(z)$ which satisfies

$$|\operatorname{tr} (\mathbf{A} \mathbf{R}_0(z)) - \operatorname{tr} (\mathbf{A} (\hat{\mathbf{K}}_0 - z)^{-1})| \rightarrow 0 \quad (33)$$

in probability as $m, d \rightarrow \infty$ (with $m/d \rightarrow c > 1$) for every bounded (in nuclear norm / sum of singular values) sequence of deterministic matrices $\mathbf{A} \in \mathbb{R}^{d \times d}$. In particular, this implies that $\frac{1}{m} \operatorname{tr} \mathbf{R}_0(z)$ approximates the Stieltjes transform of the empirical spectral measure of $\hat{\mathbf{K}}_0$. Moreover, any quadratic form with the random resolvent can be approximated by replacing it with the deterministic equivalent, which is of particular interest to us given the form of $\mathcal{F}$.

Casting $\hat{\mathbf{K}}_0 = (\mathbf{D}^{1/2} \mathbf{W})(\mathbf{D}^{1/2} \mathbf{W})^T$ as the sample covariance matrix for a dataset of d i.i.d. draws from $\mathcal{N}_m(\mathbf{0}, \mathbf{D})$, we can leverage the result of [SC95], which states that we can take

$$\mathbf{R}_0(z) = \operatorname{diag}((j^{-2\alpha} q(z) - z)^{-1} : j \in [m]), \quad (34)$$

where $q(z)$ is defined implicitly by the equation

$$q(z) := \frac{1}{1 + \frac{1}{d} \sum_{j=1}^m \frac{j^{-2\alpha}}{j^{-2\alpha} q(z) - z}}. \quad (35)$$

We verify the deterministic equivalence empirically by inspecting the alignment between the empirical spectrum of $\hat{\mathbf{K}}_0$ and the deterministic approximation (see Figure 1). As pointed out in Section 7 of [PPXP24], the latter captures the limiting eigenvalue density of $\hat{\mathbf{K}}_0$, which consists of a point mass at zero (due to the assumption $m > d$), an absolutely continuous bulk, and a sequence of spikes (referred to as “pure point outliers”) approximately corresponding to $j^{-2\alpha}$ for $j \in [d]$.

Remark 2.4. We note the contrast with the classical example of sample covariance matrices for isotropic normal random variables. The limiting spectrum of such matrices is the Marchenko-Pastur distribution [MP67], which consists solely of a point mass at zero (when $m > d$) and an absolutely continuous bulk. In our case, we also observe spikes due to the nontrivial covariance $\mathbf{D}$.

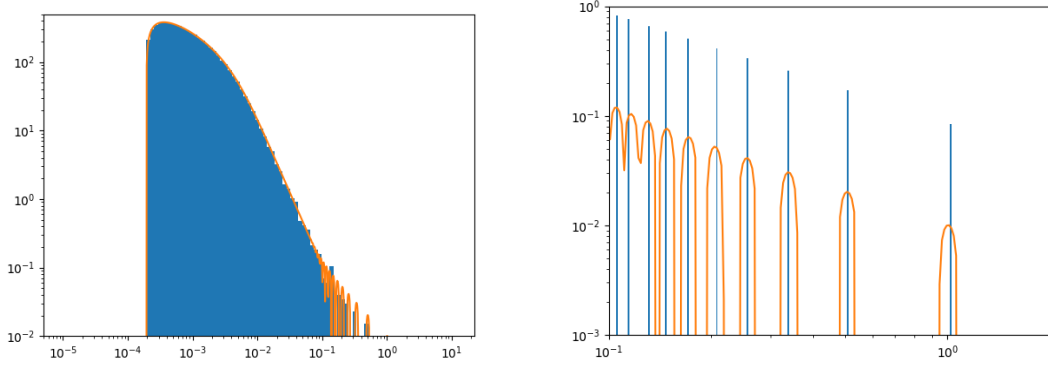


Figure 1: Spectrum of $\hat{\mathbf{K}}_0$ with $v = 2000$, $d = 1000$, $\alpha = 0.5$, and $\beta = 0.7$, averaged over 25 i.i.d. draws of $\mathbf{W}$. The deterministic approximation from Stieltjes inversion is overlaid in orange. The left plot shows the full spectrum, and the right plot is a crop displaying some of the pure point outliers.

By Stieltjes inversion (see for example [CL22] Theorem 2.1), we can approximate the limiting spectral density of $\hat{\mathbf{K}}_0$ at each $\lambda \geq 0$ via $\frac{1}{\pi} \lim_{\eta \rightarrow 0^+} \frac{1}{d} \text{tr} \mathbf{R}_0(\lambda + i\eta)$. We devise a Newton scheme which numerically solves the implicit equation for $q(z)$ and performs an approximate Stieltjes inversion. For each value of λ in our mesh, we start with $\eta = 10^4$, and at each Newton step, we logarithmically decrease η at each Newton step until we reach $\eta = 10^{-15}$. This is similar to the procedure adopted by [PPXP24] to generate Figure 6 in that paper.

Now, letting $\mathbf{a} = \mathbf{D}^{1/2}\mathbf{b}$, we define the deterministic Volterra equation by

$$\mathcal{R}(r) = \mathcal{F}(r) + (\mathcal{K} * \mathcal{R})(r), \quad (36)$$

where

$$\mathcal{F}(r) := -\frac{1}{2\pi i} \oint_{\Gamma} \mathbf{a}^T \mathbf{R}_0(z) \mathbf{a} (1 - 2\gamma Bz + 2\gamma^2 B^2 z^2)^r dz, \quad (37)$$

and

$$\mathcal{K}(r) := -\frac{1}{2\pi i} \gamma^2 B \text{tr} \left(\oint_{\Gamma} \mathbf{R}_0(z) z^2 (1 - 2\gamma Bz + 2\gamma^2 B^2 z^2)^r dz \right), \quad (38)$$

where Γ is once again any contour enclosing the spectrum of $\hat{\mathbf{K}}_0$. [PPXP24] devote their analysis to this deterministic equation and use it to characterize the risk curve over the course of training under various choices of α, β, r, d . The major drawback of this, however, is that they do not prove that the solution to this equation is indeed an approximation to the solution of the original Volterra equation (6). Instead, there is only the following conjecture.

Conjecture 2.5. *For $\{\boldsymbol{\theta}_r\}$ the sequence of iterates generated by SGD with $\boldsymbol{\theta}_0 = \mathbf{0}$ and any $\varepsilon > 0$,*

$$(1 - \varepsilon) \leq \sup_{r \in \mathbb{N}} \left\{ \frac{\mathcal{R}(\boldsymbol{\theta}_r)}{\mathcal{R}(r)} \right\} \leq (1 + \varepsilon) \quad (39)$$

with high probability (which tends to 1 as $d \rightarrow \infty$).

We provide some thoughts on proving this conjecture. It requires showing the concentration of

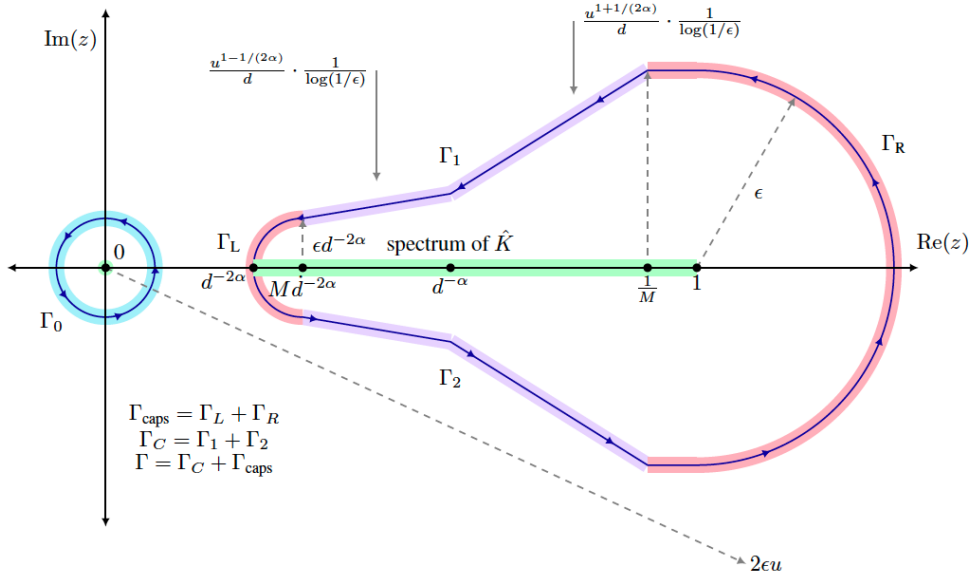


Figure 2: The contours Γ and Γ_0 . This is Figure 7 in [PPXP24].

random contour integrals. A natural first step is to show uniformity in the sense

$$\lim_{\substack{m, d \rightarrow \infty \\ \frac{m}{d} \rightarrow c}} \mathbb{P} \left(\sup_{z \in \Gamma} \left| \text{tr } \mathbf{A} \mathbf{R}_0(z) - \text{tr } \mathbf{A} (\hat{\mathbf{K}}_0 - z)^{-1} \right| > \varepsilon \right) = 0. \quad (40)$$

The problem, however, is that the specific choice of contour Γ we consider in our analysis, which is instrumental in estimating $\mathcal{R}$, is d -dependent. As d increases, it comes closer to the real axis (and thus the spectrum), where the resolvent is less “well-behaved” due to the presence of poles.

2.2 Estimating the Forcing and Kernel Functions

By Cauchy’s integral formula, instead of considering a single contour enclosing the spectrum of $\hat{\mathbf{K}}_0$, we may take a union of two contours — so as long as each point in the spectrum is enclosed by one of the two. Thus, in estimating $\mathcal{F}$ and $\mathcal{K}$, [PPXP24] consider the union of Γ_0 , a circle around the point mass at zero, and Γ , which encloses the spectral bulk. We include an illustration of this contour by the authors in Figure 2. We omit the full technical specification of Γ .

We begin with the contribution of Γ_0 . For this, we require the following proposition characterizing (and analytically extending) the implicitly defined function $q(z)$.

Proposition 2.6 (Behaviour of $q(z)$ near zero, [PPXP24] Proposition 7.3). *The function $q(z)$ is analytic in a neighbourhood of $z = 0$ of radius $c(\alpha)d^{-2\alpha}$ for some $c(\alpha) > 0$. Furthermore, q is negative on $(0, cd^{-2\alpha})$, vanishes at 0, and has $|m'(0) + \kappa(m/d)d^{2\alpha}| \leq Cd^{2\alpha-1}$ for all d sufficiently large, where $\kappa(m/d)$ solves*

$$\int_0^{m/d} \frac{\kappa}{\kappa + x^{2\alpha}} dx = 1. \quad (41)$$

Lemma 10.1 of [PPXP24] shows that (41) has a unique solution when $m > d$.

Remark 2.7. *In addition to characterizing q near zero, this result quantifies the gap between the point mass and the absolutely continuous bulk. We should take the radius of Γ_0 to be less than*

$c(\alpha)d^{-2\alpha}$ (and moreover we take Γ to enclose this point).

The following lemma is implicitly used several times in [PPXP24].

Lemma 2.8. *We have*

$$\sum_{j=1}^v j^{-c} \asymp \begin{cases} v^{1-c} & \text{if } 0 \leq c < 1 \\ 1 & \text{if } c > 1 \end{cases} = v^{(1-c)_+}. \quad (42)$$

Proof. Observe that, for $c \geq 0$,

$$\sum_{j=1}^v j^{-c} \leq \int_0^v x^{-c} dx = \frac{1}{1-c} x^{1-c} \Big|_0^v = \frac{1}{1-c} v^{1-c} \quad (43)$$

and

$$\sum_{j=1}^v j^{-c} \geq \int_1^{v+1} x^{-c} dx = \frac{1}{1-c} x^{1-c} \Big|_1^{v+1} = \frac{1}{1-c} ((v+1)^{1-c} - 1). \quad (44)$$

The result follows immediately by considering the cases $c < 1$ and $c > 1$ (which imply $1-c > 0$ and $1-c < 0$ respectively). $\square$

We are now ready to discuss the contribution of the point mass. Note that this has the natural interpretation of the approximation error, i.e., the error arising from the components of the signal and noise that the best student (for fixed $d < m$) lacks the capacity to capture.

Proposition 2.9 (Contribution along Γ_0 , [PPXP24] Proposition 8.1). *The function $\mathcal{F}_0(r)$ is constant and*

$$\left| \mathcal{F}_0(0) - \sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{1 + j^{-2\alpha} d^{2\alpha} \kappa(m/d)} \right| \lesssim C d^{-2\alpha+(2\beta-1)_+-1}. \quad (45)$$

Proof. [PPXP24] give a two-line proof sketch which we follow and flesh out in full detail here.

We have

$$\mathbf{a}^T \mathbf{R}_0(z) \mathbf{a} = \sum_{i,j=1}^m R_0(z)_{ij} a_i a_j = \sum_{j=1}^m R(z)_{jj} a_j^2 = \sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{j^{-2\alpha} q(z) - z} =: h(z), \quad (46)$$

which, in the interior of Γ_0 , has its lone singularity at $z = 0$ (since q vanishes at zero by Proposition 2.6). Hence, by the residue theorem,

$$\mathcal{F}_0(r) = -\frac{1}{2\pi i} \oint_{\Gamma_0} \sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{j^{-2\alpha} q(z) - z} (1 - 2\gamma Bz + 2\gamma^2 B^2 z^2)^r dz = -\text{res}_0 h, \quad (47)$$

which is in particular constant in r . We can calculate this residue via

$$\text{res}_0 h = \lim_{z \rightarrow 0} z h(z) = \lim_{z \rightarrow 0} \sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{\frac{1}{z} (j^{-2\alpha} q(z) - z)} = \sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{j^{-2\alpha} q'(0) - 1}. \quad (48)$$

Thus, leveraging Proposition 2.6 and 2.8,

$$\begin{aligned}
& \left| \mathcal{F}_0(0) - \sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{1 + j^{-2\alpha} d^{2\alpha} \kappa(m/d)} \right| \\
&= \left| \sum_{j=1}^m \left[\frac{j^{-2\alpha-2\beta}}{1 - j^{-2\alpha} q'(0)} - \frac{j^{-2\alpha-2\beta}}{1 + j^{-2\alpha} d^{2\alpha} \kappa(m/d)} \right] \right| \\
&= \left| \sum_{j=1}^m \left[\frac{j^{-2\alpha-2\beta} (1 + j^{-2\alpha} d^{2\alpha} \kappa(m/d)) - j^{-2\alpha-2\beta} (1 - j^{-2\alpha} q'(0))}{(1 - j^{-2\alpha} q'(0))(1 + j^{-2\alpha} d^{2\alpha} \kappa(m/d))} \right] \right| \\
&= \left| \sum_{j=1}^m \frac{j^{-4\alpha-2\beta} (q'(0) + \kappa(m/d) d^{2\alpha})}{(1 - j^{-2\alpha} q'(0))(1 + j^{-2\alpha} d^{2\alpha} \kappa(m/d))} \right| \\
&\leq C d^{2\alpha-1} \sum_{j=1}^m \frac{j^{-4\alpha-2\beta}}{(1 - C j^{-2\alpha} d^{2\alpha-1} + j^{-2\alpha} d^{2\alpha} \kappa(m/d))(1 + j^{-2\alpha} d^{2\alpha} \kappa(m/d))} \\
&\lesssim C d^{2\alpha-1} \sum_{j=1}^m \frac{j^{-4\alpha-2\beta}}{j^{-4\alpha} d^{4\alpha} \kappa(m/d)^2} \\
&= C d^{-2\alpha-1} \sum_{j=1}^m \frac{j^{-2\beta}}{\kappa(m/d)^2} \\
&\lesssim C d^{-2\alpha+(1-2\beta)_+-1}.
\end{aligned} \tag{49}$$

□

In particular, we leverage the above proposition to write

$$\mathcal{F}_0(r) = \sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{1 + j^{-2\alpha} d^{2\alpha} \kappa(m/d)} (1 + O(d^{-1})). \tag{50}$$

Indeed,

$$\sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{1 + j^{-2\alpha} d^{2\alpha} \kappa(m/d)} \asymp \frac{d^{-2\alpha}}{\kappa(m/d)} \sum_{j=1}^m j^{-2\beta} \asymp d^{-2\alpha+(1-2\beta)_+}. \tag{51}$$

We are now equipped to describe the high-dimensional asymptotics of $\mathcal{F}_0$.

Proposition 2.10 (Asymptotics of $\mathcal{F}_0$, [PPXP24] Proposition 10.3). *Assume $2\alpha + 2\beta > 1$. Then,*

$$\mathcal{F}_0(r) \asymp \begin{cases} \frac{1}{\kappa} d^{-2\alpha} \sum_{j=1}^m j^{-2\beta} & \text{if } 2\beta > 1, \\ d^{1-2\alpha-2\beta} \int_0^{m/d} \frac{u^{-2\beta}}{\kappa+u^{2\alpha}} du & \text{if } 2\beta < 1. \end{cases} \asymp d^{-2\alpha+(1-2\beta)_+}. \tag{52}$$

We leave the proof to Appendix B.

Now, we move to the contour Γ enclosing the spectral bulk. The first key result is that the contribution of the “caps” Γ_L and Γ_R is negligible compared to $\mathcal{F}_0$.

Proposition 2.11 (Contribution along Γ_{caps} , [PPXP24] Proposition 8.2). *Define*

$$\mathcal{F}_{\text{caps}} := \int_{\Gamma_L \cup \Gamma_R} \mathbf{a}^T \mathbf{R}_0(z) \mathbf{a} (1 - 2\gamma Bz + 2\gamma^2 B^2 z^2)^r dz. \tag{53}$$

There exist positive functions $f(r)$, $g(r)$ and a constant K satisfying $f(r) \leq K \exp(-c\gamma B r d^{-2\alpha})$

and $g(r) \leq K \exp(-c\gamma Br)$ so that

$$|\mathcal{F}_{caps}(r)| \leq K \cdot f(r) d^{-2\alpha+(1-2\beta)_+} + K \cdot g(r). \quad (54)$$

Hence, so long as $\gamma Br \gtrsim d^\varepsilon$ for some $\varepsilon > 0$, then $\mathcal{F}_{caps}$ is of at most the same order as $\mathcal{F}_0$. The proof of this proposition relies on the particulars of the construction of Γ and the result $|q(z) - 1| \lesssim d^{-\min\{2\alpha, 1\}}$ along Γ_R ([PPXP24] Proposition 7.6).

It remains to quantify the contribution along Γ_C .

Proposition 2.12 (Contribution along Γ_C , [PPXP24] Proposition 8.3). *Define*

$$\mathcal{F}_C(r) := \int_{\Gamma_C} \mathbf{a}^T \mathbf{R}_0(z) \mathbf{a} (1 - 2\gamma Bz + 2\gamma^2 B^2 z^2)^r dz. \quad (55)$$

There exists $M(u_0, u_1) > 0$ and a positive bounded function $C(r)$ so that if $\gamma Br \in [M, d^{2\alpha}/M]$, then

$$\frac{1}{C(r)} (\mathcal{F}_{pp}(r) + \mathcal{F}_{ac}(r)) \leq \mathcal{F}_C(r) \leq C(r) (\mathcal{F}_{pp}(r) + \mathcal{F}_{ac}(r)), \quad (56)$$

where

$$\mathcal{F}_{pp}(r) := \frac{1}{2\alpha} \int_0^1 u^{(2\beta-1)/2\alpha} \exp(-2\gamma Bru) du \quad (57)$$

and

$$\mathcal{F}_{ac}(r) := \frac{c_\beta}{2\alpha} \int_{d^{-2\alpha}}^1 u^{-1/2\alpha} d^{-1} \exp(-2\gamma Bru) du, \quad (58)$$

with $c_\beta = \sum_{j=1}^\infty j^{-2\beta}$ if $\beta > 1/2$ and $c_\beta = 0$ otherwise.

$\mathcal{F}_{pp}$ captures the contribution of the spike eigenvalues, while $\mathcal{F}_{ac}$ captures the absolutely continuous spectral bulk. The proof of the above proposition and the verification of this interpretation requires several technical estimates of $q(z)$ along Γ_C (see in particular [PPXP24] Propositions 7.4 and 7.5). It is then followed up by an asymptotic analysis of $\mathcal{F}_{pp}$ and $\mathcal{F}_{ac}$.

Proposition 2.13 (Asymptotics of $\mathcal{F}_{pp}$, [PPXP24] Proposition 10.2). *Assume $2\alpha + 2\beta > 1$. For all $\varepsilon > 0$, there exists an $M > 0$ such that if $\gamma Br > M$, then*

$$(1 - \varepsilon)g(r) \leq \mathcal{F}_{pp}(r) \leq (1 + \varepsilon)g(r), \quad (59)$$

where

$$g(r) := (2\alpha)^{-1} (2\gamma B)^{\frac{1}{2\alpha} - \frac{\beta}{\alpha} - 1} \Gamma\left(\frac{\beta}{\alpha} - \frac{1}{2\alpha} + 1\right) r^{-(1 + \frac{\beta}{\alpha}) + \frac{1}{2\alpha}}. \quad (60)$$

Here, Γ denotes the Gamma function.

Proposition 2.14 (Asymptotics of $\mathcal{F}_{ac}$, [PPXP24] Proposition 10.4). *There exists a constant $K(\alpha, \beta) > 0$ such that*

$$\mathcal{F}_{ac}(r) \leq \begin{cases} K\mathcal{F}_0(r) & \text{if } 2\beta > 1, 2\alpha < 1 \\ 0 & \text{if } 2\beta < 1. \end{cases} \quad (61)$$

On the other hand, if $2\alpha > 1$ and $2\beta > 1$, we have the following: for all $\varepsilon > 0$ there exists $M > 0$ such that if $\gamma Br \in [M, d^{2\alpha}/M]$,

$$(1 - \varepsilon)g(r) \leq \mathcal{F}_{ac}(r) \leq (1 + \varepsilon)g(r), \quad (62)$$

where

$$g(r) := \left(\sum_{j=1}^m j^{-2\beta} \right) (2\gamma B)^{-1+\frac{1}{2\alpha}} (2\alpha)^{-1} \Gamma(1 - \frac{1}{2\alpha}) r^{-1+\frac{1}{2\alpha}} d^{-1}. \quad (63)$$

Moving on to the kernel function $\mathcal{K}$, we can quickly see from the lemma below that the point mass at zero does not contribute.

Lemma 2.15 (Trace of $\mathbf{R}_0$, [PPXP24] Lemma 7.1). *For any $z \in \mathbb{C}$ such that $\text{Im}(q(z)) > 0$, we have the identity*

$$\sum_{j=1}^m \frac{1}{j^{-2\alpha} q(z) - z} = \frac{-m + (1 - q(z))d}{z}. \quad (64)$$

With this, the contribution of Γ_0 to $\mathcal{K}$ becomes

$$\mathcal{K}_0(r) := -\gamma^2 B \frac{1}{2\pi} \oint_{\Gamma_0} [-m + (1 - q(z))d] z (1 - 2\gamma Bz + 2\gamma^2 B^2 z^2)^r dz, \quad (65)$$

which is zero by analyticity of q and the residue theorem.

In fact, $\mathcal{K}$ is well-approximated by the contribution from Γ_C alone.

Proposition 2.16 (Approximation of $\mathcal{K}$, [PPXP24] Proposition 9.1). *Suppose $\alpha > 1/4$. Then there exists a positive function $C(r)$ such that*

$$\frac{1}{C(r)} \mathcal{K}_{pp}(r) \leq \mathcal{K}(r) \leq C(r) \mathcal{K}_{pp}(r), \quad (66)$$

where

$$\mathcal{K}_{pp}(r) := \frac{\gamma^2 B}{2\alpha} \int_0^1 u^{1-\frac{1}{2\alpha}} \exp(-2\gamma B r u) du. \quad (67)$$

$C(r)$ is bounded independent of d if $\gamma B r < d^{2\alpha} M$ for some $M > 0$.

The asymptotic analysis of $\mathcal{K}_{pp}(r)$ is similar to the one for $\mathcal{F}_{pp}$.

Proposition 2.17 (Asymptotics of $\mathcal{K}_{pp}$, [PPXP24] Proposition 10.5). *Suppose $\alpha > 1/4$. For any $\varepsilon > 0$, there exists $M > 0$ such that if $\gamma B r \geq M$, then*

$$(1 - \varepsilon)g(r) \leq \mathcal{K}_{pp}(r) \leq (1 + \varepsilon)g(r), \quad (68)$$

where

$$g(r) := (2\alpha)^{-1} \gamma^2 B (2\gamma B)^{-2+\frac{1}{2\alpha}} \Gamma(2 - \frac{1}{2\alpha}) r^{-2+\frac{1}{2\alpha}}. \quad (69)$$

2.3 The Phases of Scaling Laws

It now comes to plug these bounds back into the deterministic Volterra equation (36). [PPXP24] prove that the term $(\mathcal{K} * \mathcal{P})(r)$ can be approximated so that $\mathcal{P}(r)$ can be explicitly bounded from above and below. Crucially, this requires an additional assumption on the step size γ and batch size B .

Theorem 2.18 (Approximation Solution for $\mathcal{P}$, [PPXP24] Theorem 2.1). *Suppose that γ, B are such that $\gamma < 1/2$ and $1/4(1 - \sqrt{1 - \frac{4}{B}}) > \gamma$. Moreover assume $2\alpha + 2\beta > 1$ and $\alpha > 1/4$.⁵ There*

⁵The authors conjecture that this result also holds for $\alpha \leq 1/4$.

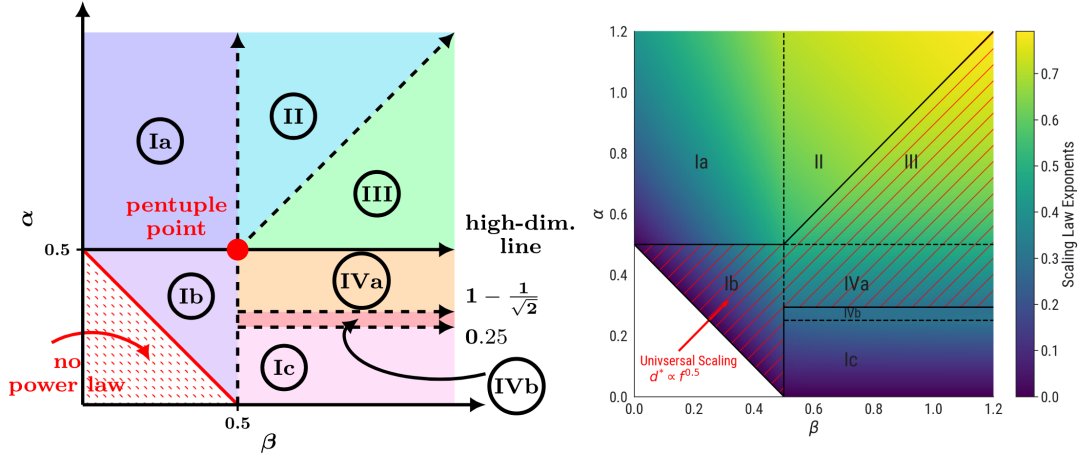


Figure 3: Illustration of the 4(+3) phases in [PPXP24]. This is Figure 3 from their paper. The lower boundary of Phase Ib (shown in red) is the line $2\alpha + 2\beta = 1$. The boundary between Phases II and III is the line $\alpha = \beta$.

exist $M > 0$ and $K = K(\alpha, \beta, M)$ such that if $\gamma Br > M$, then we can explicitly bound $\mathcal{P}$

$$\mathcal{F}(r) + (\mathcal{K} * \mathcal{F})(r) \leq \mathcal{R}(r) \leq \mathcal{F}(r) + K(\mathcal{K} * \mathcal{F})(r). \quad (70)$$

Moreover, the convolution can be bounded as

$$(\mathcal{K} * \mathcal{F})(r) \asymp \mathcal{F}(r) + \frac{1}{\gamma B} \mathcal{K}(r). \quad (71)$$

Hence, for γBr satisfying the conditions of the above results, we may write

$$\mathcal{R}(r) \asymp \mathcal{F}_0(r) + \mathcal{F}_{pp}(r) + \mathcal{F}_{ac}(r) + \frac{1}{\gamma B} \mathcal{K}_{pp}(r). \quad (72)$$

Thus, characterizing the scaling behaviour of the risk given α, β reduces to determining which of these terms asymptotically dominates. This gives a partition of the first quadrant of the (α, β) plane into what [PPXP24] refer to as “phases”. Figure 3 illustrates this partition, while Figure 4 summarizes the learning curve under compute-optimal scaling in each phase. The former figure highlights the intuitive finding that for “easier tasks” (i.e., large α and β), the risk decays faster with C (assuming that the choice of d is compute-optimal).

The procedure for characterizing the dominant term in each phase is essentially the same. We focus on Phase Ia for the sake of illustration.

Proposition 2.19 (Dominant Contributions in Phase Ia, [PPXP24] Proposition 5.3). *Suppose $2\beta < 1$, $2\alpha > 1$, and that γ, B satisfy the conditions of Theorem 2.18. There exists $M > 0$ such that if $\gamma Br > M$, then*

$$\mathcal{R}(r) \asymp \mathcal{F}_0(r) + \mathcal{F}_{pp}(r). \quad (73)$$

Proof. Since $2\beta < 1$, $\mathcal{F}_{ac}(r) = 0$ by Proposition 2.14. Moreover, $\mathcal{F}_{pp} \asymp r^{-(1+\frac{\beta}{\alpha})+\frac{1}{2\alpha}}$ by Proposition 2.13 and $\mathcal{K}_{pp} \asymp r^{-2+\frac{1}{2\alpha}}$ by Proposition 2.17. Comparing the latter two, we have

$$\frac{\mathcal{F}_{pp}}{\mathcal{K}_{pp}} \asymp r^{2-(1+\frac{\beta}{\alpha})} = r^\varepsilon \text{ for some } \varepsilon > 0 \quad (74)$$

	Loss $\mathcal{P}(r)$	Trade off	Compute-optimal Curves
Phase I	$\mathcal{F}_{pp}(r) + \mathcal{F}_0(r)$	$\mathcal{F}_{pp} = \mathcal{F}_0$	Ia $\tilde{\mathcal{P}}_{\text{Phase Ia}}^*(f) \asymp f^{\left(\frac{1}{2\alpha+1}-1\right)(1+\beta/\alpha-1/(2\alpha))}$ $d_{\text{Phase Ia}}^* \asymp f^{1/(2\alpha+1)}$
			Ib $\tilde{\mathcal{P}}_{\text{Phase Ib}}^*(f) \asymp f^{\frac{1}{2}-\alpha-\beta}$ $d_{\text{Phase Ib}}^* \asymp f^{\frac{1}{2}}$
			Ic $\tilde{\mathcal{P}}_{\text{Phase Ic}}^*(f) \asymp f^{\frac{\alpha(2\alpha+2\beta-1)}{\alpha(2\beta-3)-2\beta+1}}$ $d_{\text{Phase Ic}}^* \asymp f^{\frac{1-2(\alpha+\beta)}{2(\alpha(2\beta-3)-2\beta+1)}}$
Phase II	$\mathcal{F}_{pp}(r) + \mathcal{F}_{ac}(r) + \mathcal{F}_0(r)$	$\mathcal{F}_{pp} = \mathcal{F}_{ac}$	$\tilde{\mathcal{P}}_{\text{Phase II}}^*(f) \asymp f^{-\frac{2\alpha+2\beta-1}{2(\alpha+\beta)}}$ $d_{\text{Phase II}}^* \asymp f^{(\beta/\alpha)/(1+\beta/\alpha)}$
Phase III	$\mathcal{F}_{ac}(r) + \mathcal{F}_0(r) + \frac{1}{\gamma B} \mathcal{K}_{pp}(r)$	$\frac{1}{\gamma B} \mathcal{K}_{pp} = \mathcal{F}_{ac}$	$\tilde{\mathcal{P}}_{\text{Phase III}}^*(f) \asymp f^{(1-4\alpha)/(4\alpha)}$ $d_{\text{Phase III}}^* \asymp f^{1/2}$
Phase IV	$\mathcal{F}_{pp}(r) + \mathcal{F}_0(r) + \frac{1}{\gamma B} \mathcal{K}_{pp}(r)$	IVa $\frac{1}{\gamma B} \mathcal{K}_{pp} = \mathcal{F}_0$	$\tilde{\mathcal{P}}_{\text{Phase IVa}}^*(f) \asymp f^{-\alpha}$ $d_{\text{Phase IVa}}^* \asymp f^{1/2}$
		IVb $\frac{1}{\gamma B} \mathcal{K}_{pp} = \mathcal{F}_{pp}$	$\tilde{\mathcal{P}}_{\text{Phase IVb}}^*(f) \asymp f^{\frac{(1-2\alpha)(2\alpha+2\beta-1)}{(2(2\alpha\beta+\alpha-2\beta))}}$ $d_{\text{Phase IVb}}^* \asymp f^{(\alpha-\beta)/(2\alpha\beta+\alpha-2\beta)}$

Figure 4: Summary table of the 4(+3) phases in [PPXP24]. This is Table 2 from their paper. In their notation, $\tilde{\mathcal{P}}^*$ denotes the risk $\mathcal{R}$ under the compute-optimal choice of d , and f is the number of flops C .

since $\beta/\alpha < 1$. Hence, $\mathcal{F}_{pp}$ is of strictly higher order than $\frac{1}{\gamma B} \mathcal{K}_{pp}$ (note here that γB is of constant order). $\square$

Proposition 2.20 (Compute-Optimality in Phase Ia, [PPXP24] Proposition 6.1). *Consider the same assumptions as in Proposition 2.19. Then the compute-optimal choice of d is of order $C^{\frac{1}{2\alpha+1}}$ and the associated risk is*

$$\mathcal{R}^*(C) \asymp C^{\left(\frac{1}{2\alpha+1}-1\right)(1+\frac{\beta}{\alpha}-\frac{1}{2\alpha})}. \quad (75)$$

Proof. From Proposition 2.19, we have (abusing notation)

$$\mathcal{R}(r, d) \asymp \mathcal{F}_0(r) + \mathcal{F}_{pp}(r) \asymp d^{1-2\alpha-2\beta} + r^{\frac{1}{2\alpha}-\frac{\beta}{\alpha}-1}. \quad (76)$$

Since batch size B is of constant order, we have $C \asymp rd$, and so compute-optimality is achieved when

$$d^{1-2\alpha-2\beta} = \left(\frac{C}{d}\right)^{\frac{1}{2\alpha}-\frac{\beta}{\alpha}-1}. \quad (77)$$

Solving for d , we have

$$d^* \asymp C^{\frac{\frac{1}{2\alpha}-\frac{\beta}{\alpha}-1}{\frac{1}{2\alpha}-2\alpha-2\beta-\frac{\beta}{\alpha}}}. \quad (78)$$

Notice that

$$\left(\frac{1}{2\alpha} - \frac{\beta}{\alpha} - 1\right)(2\alpha + 1) = \frac{1}{2\alpha} - 2\beta - \frac{\beta}{\alpha} - 2\alpha,$$

and thus $d^* \asymp C^{\frac{1}{2\alpha+1}}$ as desired. The form of the associated risk follows immediately from $r^* \asymp C^{\frac{2\alpha}{2\alpha+1}}$ and (75). $\square$

3 Extension to Spiked Model

As we have discussed, the bulk of the theoretical scaling laws literature is centred on the linear random features setting. In this report, we propose an incremental extension in the direction of *feature learning* [YH21, BES⁺22, CPD⁺24]. By this, we mean that we wish to simulate the ability of neural networks to learn vector representations of the input data in their intermediate layers. In practice, this is achieved by training all layers of the network simultaneously.

Our simplified setting is inspired by a result from [BES⁺22] on the intermediate layer parameters of a two-layer neural network after a single step of gradient descent.

Theorem 3.1 (Spike in First-Layer Weights after One GD Step, [BES⁺22] Theorem 3). *Assume that the student network takes the form $f_{NN}(\mathbf{x}) = \langle \sigma(\mathbf{W}^T \mathbf{x}), \boldsymbol{\theta} \rangle$, where σ is a nonlinear activation function with bounded first three derivatives almost surely, $\mathbb{E}[\sigma(Z)] = 0$, and $\mu_1 := \mathbb{E}[Z\sigma(Z)] \neq 0$ for $Z \sim \mathcal{N}(0, 1)$. Moreover, $\mathbf{W}$ is initialized with i.i.d. $\mathcal{N}(0, 1/m)$ entries and $\boldsymbol{\theta}$ with i.i.d. $\mathcal{N}(0, 1/d)$ entries. Write the teacher f^* as*

$$f^*(\mathbf{x}) = \mu_0^* + \mu_1^* \langle \mathbf{x}, \mathbf{b}_* \rangle + P_{>1} f^*(\mathbf{x}), \quad (79)$$

where $P_{>1}$ is the projector orthogonal to constant and linear functions in $L^2(\mathbb{R}^d, \mathcal{N}(0, 1))$.

Assume data is generated as $y_i = f^*(\mathbf{x}_i) + \varepsilon_i$, where $x \sim \mathcal{N}_m(\mathbf{0}, \mathbf{I}_m)$ and ε_i is mean-zero subgaussian. Let n be the dataset size and assume $\mathbf{W}_1$ is the result of a single full-batch gradient descent step with learning rate $\gamma = \Theta(1)$ on $\mathbf{W}$. Consider the proportional asymptotic regime $n, d, m \rightarrow \infty$ with $n/m \rightarrow \psi_1 \in (0, \infty)$, $d/m \rightarrow \psi_2 \in (0, \infty)$. Define $\bar{\mu} = \lim_{d \rightarrow \infty} \|f^*\|_{L^2(\mathbb{R}^d, \mathcal{N}(0, 1))}$, and

$$\theta_1 := \sqrt{\bar{\mu}^2 \psi_1^{-1} + \mu_1^{*2}} \cdot \mu_1 \gamma, \quad \theta_2 := \mu_1 \mu_1^* \gamma. \quad (80)$$

Then the leading singular value $s_1(\mathbf{W}_1)$ and the corresponding left singular vector $\mathbf{u}_1$ satisfy

$$s_1(\mathbf{W}_1) \rightarrow \sqrt{\frac{(1 + \theta_1^2)(\psi_2 + \theta_1^2)}{\theta_1^2}}, \quad |\langle \mathbf{u}_1, \mathbf{b} \rangle|^2 \rightarrow \frac{\theta_2^2}{\theta_1^2} \left(1 - \frac{\psi_2 + \theta_1^2}{\theta_1^2(\theta_1^2 + 1)} \right), \quad (81)$$

if $\theta_1 > \psi_2^{1/4}$; otherwise $s_1(\mathbf{W}_1) \xrightarrow{\mathbb{P}} 1 + \sqrt{\psi_2}$ and $|\langle \mathbf{u}_1, \mathbf{b} \rangle| \xrightarrow{\mathbb{P}} 0$. Moreover, for any $i > 1$, we have $|s_i(\mathbf{W}_i) - s_i(\mathbf{W}_0)| \xrightarrow{\mathbb{P}} 0$.

This result states that if the size μ_1^* of the linear component of the teacher (or the step size γ) is sufficiently large, then (in the asymptotic regime described in the theorem's assumptions), there will be a spike in the spectrum of $\mathbf{W}_1 \mathbf{W}_1^T$ having nontrivial alignment with the ground truth direction $\mathbf{b}$. Moreover, the alignment improves as μ_1^* and γ increase. The remainder of the spectrum of $\mathbf{W}_1 \mathbf{W}_1^T$ (i.e., the bulk) is unaffected.

Now, of course, the modelling setup assumed in the above theorem differs from our linear random features setup. In particular, the input is assumed isotropic and the student can learn nonlinear functions of it. A step of full-batch gradient descent is performed, as opposed to mini-batch SGD. However, it still provides a window into learning dynamics in nonlinear deep neural

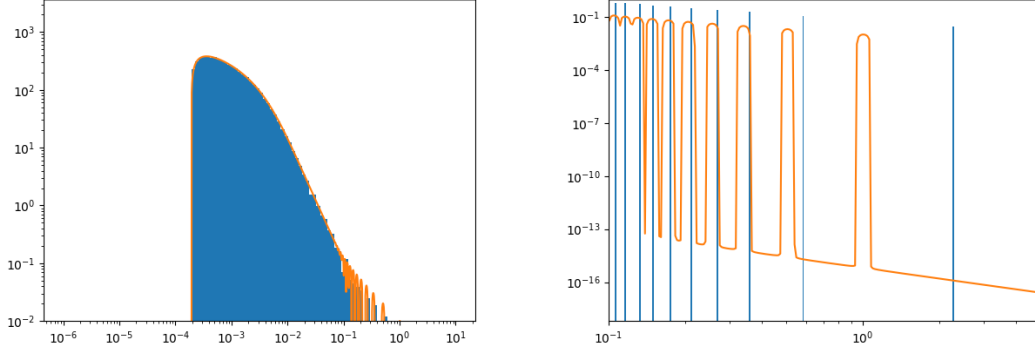


Figure 5: Empirical spectrum of $\hat{\mathbf{K}}$ for $\alpha = 0.5$, $\beta = 0.7$, $v = 2000$, $d = 1000$, $\tau = 1/\sqrt{d}$. This is compared to the continuous approximation to the spectrum with $\tau = 0$ (in orange) from Figure 1. The left-hand plot shows the full spectrum, while the right-hand plot shows the spikes between 0.1 and 10.

networks, which must be characterized in order to develop a theory of scaling laws for such models which are closer to practice. We take this as a starting point.

Hence, the extension we make to the [PPXP24] framework is to introduce a spike in the spectrum of $\mathbf{W}$ in the direction of the signal $\mathbf{b}$. We assume

$$\mathbf{W} = \tau \mathbf{b} \mathbf{1}^T + \mathbf{Z}, \quad (82)$$

where $\tau > 0$ is fixed, $\mathbf{1}$ denotes the vector of ones in $\mathbb{R}^d$, and $\mathbf{Z} \in \mathbb{R}^{v \times d}$ is a matrix with i.i.d. $\mathcal{N}(0, 1/d)$ entries. $\mathbf{b}$ is the same signal vector as in the previous section. Since the only change from [PPXP24] is to $\mathbf{W}$, and the entire analysis in Section 2 is conditional on $\mathbf{W}$, we only require a deterministic equivalent for our new $\hat{\mathbf{K}}_0 = \mathbf{D}^{1/2} \mathbf{W} \mathbf{W}^T \mathbf{D}^{1/2}$.

3.1 Deterministic Equivalent for $\hat{\mathbf{K}}$

Notice that the entries of $\mathbf{W}$ are no longer i.i.d., so we cannot immediately use classical results for sample covariance matrices from [SC95] to obtain a deterministic equivalent. Instead, $\hat{\mathbf{K}}$ has the interpretation of a (non-centred) sample covariance matrix for a dataset of d i.i.d. draws from $\mathcal{N}_m(\tau \mathbf{D}^{1/2} \mathbf{b}, \mathbf{D})$. We illustrate the resulting empirical spectrum in Figure 5. While the absolutely continuous bulk is unchanged — as would be expected from the theory of spiked sample covariance matrices (e.g., [BBAP05]) — the pure point outliers from $\hat{\mathbf{K}}_0$ have “shifted to the right”.

Remark 3.2. *We contrast this with rank-one spike models in the isotropic case, where a single spike eigenvalue can be observed to the right of the bulk if the rank-one perturbation is sufficiently large in magnitude. This is the well-known BBP phase transition phenomenon [BBAP05].*

Our objective is to find a deterministic equivalent for $\hat{\mathbf{K}}$ in terms of $\mathbf{R}_0$, since this would allow us to leverage all of the results on $q(z)$ — which necessitate lengthy proofs using complex analysis tools — in [PPXP24], some of which we covered in Section 2. It is natural to expand

$$\begin{aligned} \hat{\mathbf{K}} &= \mathbf{D}^{1/2} \mathbf{W} \mathbf{W}^T \mathbf{D}^{1/2} = \mathbf{D}^{1/2} (\tau \mathbf{b} \mathbf{1}^T + \mathbf{Z}) (\tau \mathbf{b} \mathbf{1}^T + \mathbf{Z})^T \mathbf{D}^{1/2} \\ &= \tau^2 d \mathbf{a} \mathbf{a}^T + \tau \mathbf{a} \mathbf{1}^T \mathbf{Z}^T \mathbf{D}^{1/2} + \tau \mathbf{D}^{1/2} \mathbf{Z} \mathbf{1} \mathbf{a}^T + \mathbf{D}^{1/2} \mathbf{Z} \mathbf{Z}^T \mathbf{D}^{1/2}. \end{aligned} \quad (83)$$

We show in Appendix C that when $\tau = \Theta(1/\sqrt{d})$, the first and last term are of constant order, while the cross terms vanish at a $1/\sqrt{d}$ rate in operator norm. This suggests that we can easily connect $(\hat{\mathbf{K}} - z)^{-1}$ to $(\hat{\mathbf{K}}_0 - z)^{-1}$ with the Sherman-Morrison formula. However, this approach fails as $(\hat{\mathbf{K}}_0 + \tau^2 d \mathbf{a} \mathbf{a}^T - z)^{-1}$ and $(\hat{\mathbf{K}}_0 - z)^{-1}$ have the same deterministic equivalent $\mathbf{R}_0$. As we have seen in Figure 5, $\mathbf{R}_0$ does not capture the rightward shift in the spike eigenvalues.

Instead, we consider a more flexible approach to deriving deterministic equivalents from [HLN07]. Adapting our notation to fit their formalism, we decompose

$$\underbrace{\mathbf{D}^{1/2} \mathbf{W}}_{\Sigma} = \underbrace{\tau \mathbf{a} \mathbf{1}^T}_{\mathbf{A}} + \underbrace{\mathbf{D}^{1/2} \mathbf{Z}}_{\mathbf{V}}, \quad (84)$$

$\mathbf{V}$ has the form

$$V_{ij} = \frac{\sigma_{ij}}{\sqrt{d}} U_{ij}, \quad (85)$$

where $\sigma_{ij}^2 = i^{-2a}$ and $U_{ij} \sim \mathcal{N}(0, 1)$.

By Theorems 2.4 and 2.5 of [HLN07], the deterministic equivalents $\mathbf{R}(z)$ of $(\hat{\mathbf{K}} - z)^{-1}$ and $\tilde{\mathbf{R}}(z)$ of $(\mathbf{W}^T \mathbf{D} \mathbf{W} - z)^{-1}$ are characterized by the system of equations

$$\psi_i(z) = -\frac{1}{z(1 + \frac{1}{d} \text{tr} \mathbf{a}_i \tilde{\mathbf{R}}(z))} \quad \text{for } 1 \leq i \leq m, \quad (86)$$

$$\tilde{\psi}_j(z) = -\frac{1}{z(1 + \frac{1}{d} \text{tr} \mathbf{B}_j \mathbf{R}(z))} \quad \text{for } 1 \leq j \leq d, \quad (87)$$

$$\mathbf{R}(z) = (\Psi^{-1} - z \mathbf{A} \tilde{\Psi}(z) \mathbf{A}^T)^{-1} \quad (88)$$

$$\tilde{\mathbf{R}}(z) = (\tilde{\Psi}^{-1} - z \mathbf{A}^T \Psi(z) \mathbf{A})^{-1}, \quad (89)$$

where $\mathbf{B}_j = \text{diag}(\sigma_{ij}^2 : 1 \leq i \leq m)$, $\mathbf{a}_i = \text{diag}(\sigma_{ij}^2 : 1 \leq j \leq d)$, and Ψ and $\tilde{\Psi}$ are defined as diagonal matrices in the obvious way.

Substituting, we have

$$\Psi = -\frac{1}{z} (\mathbf{I}_v + \frac{1}{d} \mathbf{D} \text{tr} \tilde{\mathbf{R}})^{-1}, \quad (90)$$

$$\tilde{\Psi} = -\frac{1}{z} (1 + \frac{1}{d} \text{tr} \mathbf{D} \mathbf{R})^{-1} \mathbf{I}_d. \quad (91)$$

Then, letting $s(z) = \text{tr} \tilde{\mathbf{R}}(z)$, and $t(z) = \text{tr} \mathbf{D} \mathbf{R}(z)$, we have that the deterministic equivalent is

$$\begin{aligned} \mathbf{R} &= (\Psi^{-1} - z \mathbf{A} \tilde{\Psi} \mathbf{A}^T)^{-1} \\ &= (-z(\mathbf{I}_v + \frac{s}{d} \mathbf{D}) + z\tau^2 d \mathbf{a} \mathbf{1}^T \frac{1}{z}(1 + \frac{t}{d})^{-1} \mathbf{1} \mathbf{a}^T)^{-1} \\ &= (-z(\mathbf{I}_v + \frac{s}{d} \mathbf{D}) + \tau^2 d (1 + \frac{t}{d})^{-1} \mathbf{a} \mathbf{a}^T)^{-1} \\ &= -\frac{1}{z} (\mathbf{I}_v + \frac{s}{d} \mathbf{D})^{-1} - \frac{\frac{\tau^2 d}{z^2} (1 + \frac{t}{d})^{-1} (\mathbf{I}_v + \frac{s}{d} \mathbf{D})^{-1} \mathbf{a} \mathbf{a}^T (\mathbf{I}_v + \frac{s}{d} \mathbf{D})^{-1}}{1 - \frac{\tau^2 d}{z} (1 + \frac{t}{d})^{-1} \mathbf{a}^T (\mathbf{I}_v + \frac{s}{d} \mathbf{D})^{-1} \mathbf{a}}, \end{aligned} \quad (92)$$

where we used the Sherman-Morrison formula in the last line.

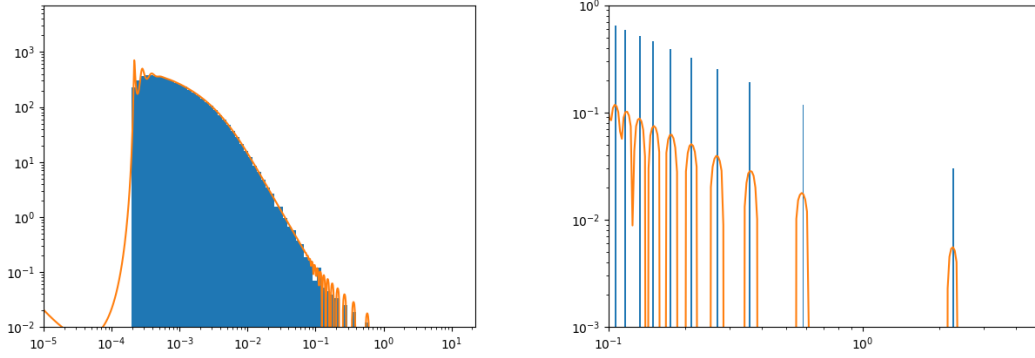


Figure 6: Deterministic approximation (orange) by the method of [HLN07] to the empirical spectrum (blue) of $\hat{\mathbf{K}}$ for $\alpha = 0.5$, $\beta = 0.7$, $v = 2000$, $d = 1000$, $\tau = 1/\sqrt{d}$. The left-hand plot shows the full spectrum, while the right-hand plot shows the spikes between 0.1 and 10.

Moreover,

$$\begin{aligned}
\tilde{\mathbf{R}} &= (\tilde{\Psi}^{-1} - z\mathbf{A}^T\tilde{\Psi}\mathbf{A})^{-1} \\
&= \left(-z\left(1 + \frac{t}{d}\right)\mathbf{I}_d + z\tau^2\mathbf{1a}^T\frac{1}{z}(\mathbf{I}_v + \frac{s}{d}\mathbf{D})^{-1}\mathbf{a}\mathbf{1}^T\right)^{-1} \\
&= \left(-z\left(1 + \frac{t}{d}\right)\mathbf{I}_d + \tau^2\mathbf{1a}^T(\mathbf{I}_v + \frac{s}{d}\mathbf{D})^{-1}\mathbf{a}\mathbf{1}^T\right)^{-1} \\
&= -\frac{1}{z}\left(1 + \frac{t}{d}\right)^{-1}\mathbf{I}_d - \frac{\frac{\tau^2}{z^2}\left(1 + \frac{t}{d}\right)^{-2}(\mathbf{1a}^T(\mathbf{I}_v + \frac{s}{d}\mathbf{D})^{-1}\mathbf{a}\mathbf{1}^T)}{1 - \frac{\tau^2 d}{z}\left(1 + \frac{t}{d}\right)^{-1}\mathbf{a}^T(\mathbf{I}_v + \frac{s}{d}\mathbf{D})^{-1}\mathbf{a}},
\end{aligned} \tag{93}$$

where we have once again used the Sherman-Morrison formula in the last line.

To simplify the notation, let $\mathbf{S} = (\mathbf{I}_v + \frac{s}{d}\mathbf{D})^{-1}$ and $u = (1 + \frac{t}{d})^{-1}$. From the above, we can set up a system of scalar equations for s and t .

$$f_1(s, t) := s + \frac{d}{z}u + \frac{\frac{\tau^2}{z^2}du^2\mathbf{a}^T\mathbf{S}\mathbf{a}}{1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a}} = 0 \tag{94}$$

$$f_2(s, t) := t + \text{tr } \mathbf{D} \left(\frac{1}{z}\mathbf{S} + \frac{\frac{\tau^2}{z^2}du\mathbf{S}\mathbf{a}\mathbf{a}^T\mathbf{S}}{1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a}} \right) = t + \frac{1}{z}\text{tr } \mathbf{D}\mathbf{S} + \frac{\frac{\tau^2}{z^2}du\mathbf{a}^T\mathbf{D}\mathbf{S}^2\mathbf{a}}{1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a}} = 0. \tag{95}$$

We solve the above system with our Newton scheme described in Section 2 while logarithmically decreasing the imaginary part of z to 10^{-15} . We detail the Jacobian computation for Newton's method in Appendix D. In Figure 6, we observe that the deterministic equivalent $\mathbf{R}$ computed from this method exhibits solid alignment with the empirical spectrum for $\tau = 1/\sqrt{d}$.⁶

Unfortunately, as alluded to earlier, this deterministic equivalent is not in terms of q and requires a fresh analysis of the implicit equations that define it. We leave this analysis (or, preferably, the derivation of a simpler deterministic equivalent) to future work.

3.2 Simulation Study

We perform an experiment to investigate the hypothesis that a rank-one perturbation in the direction of the signal leads to a “better” scaling law. We follow the experimental setup in [PPXP24].

⁶It should be noted that there appears to be some instability on the left end of the bulk, perhaps due to numerical errors.

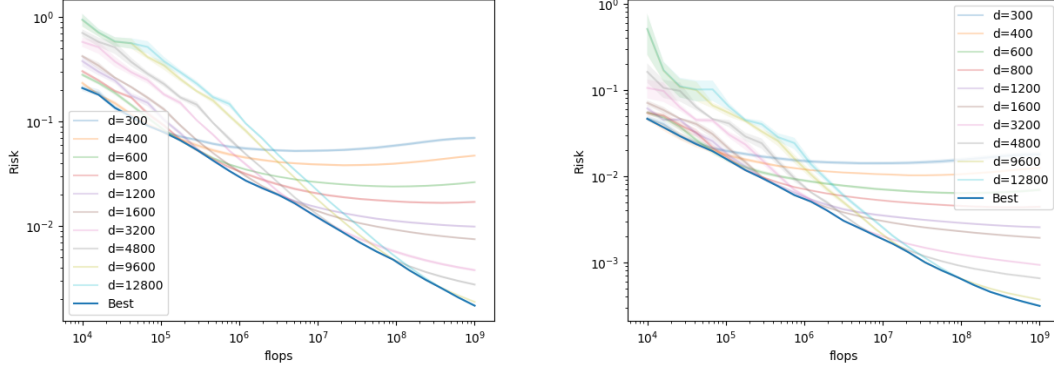


Figure 7: Results of scaling simulation for $(\alpha, \beta) = (0.5, 0.7)$. Learning curves for each choice of d are shown, allowing with the best curve (taking the minimum risk over d at each mesh point). The case $\tau = 0$ is on the left and the case $\tau = 1/\sqrt{d}$ is on the right.

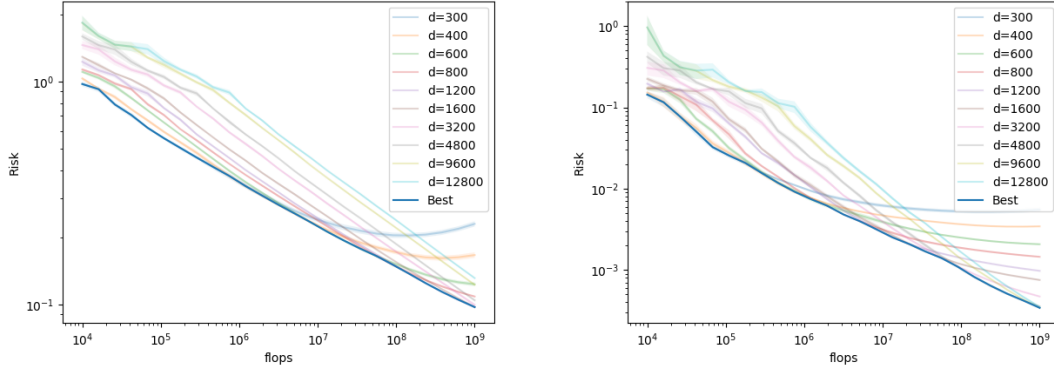


Figure 8: Results of scaling simulation for $(\alpha, \beta) = (0.6, 0.2)$. Learning curves for each choice of d are shown, allowing with the best curve (taking the minimum risk over d at each mesh point). The case $\tau = 0$ is on the left and the case $\tau = 1/\sqrt{d}$ is on the right.

All code for this subsection is included in the supplementary material. The implementation is in Python using the JAX library [FJL18] to leverage GPU computation.

For this report, we focus on the case $(\alpha, \beta) = (0.5, 0.7)$, which is at the boundary between Phases III and IVa, and $(\alpha, \beta) = (0.6, 0.2)$. We set $v = 14,080$ and consider training with various choices of d (300, 400, 600, 800, 1200, 1600, 2400, 3200, 4800, 6400, 9600, 12800). We consider different choices of step size γ (0.01, 0.1, 0.2, 0.5) but consistently find 0.1 to work best. However, we do not tune γ for each individual choice of d and instead hold it constant (as in the theory). Batch size is fixed at 1. Leveraging the proof of Proposition 2.1, the risk is calculated exactly as

$$\mathcal{R}_d(\theta_r) = \|\mathbf{D}^{1/2}(\mathbf{W}\theta_r - \mathbf{b})\|_2^2. \quad (96)$$

We plot the results on a logarithmically spaced mesh with 25 points between $C = 10^4$ and $C = 10^9$. At each mesh point C_k , we iterate over all choices of d and run SGD for $\lfloor C_k/d \rfloor$ iterations. We save the best (i.e., lowest) risk achieved among all these choices. We then fit a power law of the form $\hat{\mathcal{R}}(C) = sC^{-\eta}$ by OLS using the data from the mesh points. We compare the power law fits for $\tau = 0$ and $\tau = 1/\sqrt{d}$. We display the results in Figures 7 and 8.

For the case $(\alpha, \beta) = (0.5, 0.7)$ with $\tau = 0$, we obtain the fitted power law coefficients $s = 9.862$, $\eta = 0.4175$, compared to the theoretical scaling exponent of 0.5 (from Phases III and IVa in Figure 4). For $\tau = 1/\sqrt{d}$, OLS gives $s = 2.756$ and $\eta = 0.4503$, suggesting that the addition of the rank-one perturbation leads to faster decay of the risk. The effect is even more pronounced for $(\alpha, \beta) = (0.6, 0.2)$. In this case, when $\tau = 0$, our fitted power law has $\eta = 0.2002$ (compared to a theoretical value of 0.2727). Meanwhile, for $\tau = 1/\sqrt{d}$, we obtain a much larger exponent $\eta = 0.5058$. This is consistent with what was observed by [BAP24a] in their linear feature learning setting. The smaller the β , the harder the task (as the number of components of the signal $\mathbf{b}$ exceeding a given threshold increases), and the more helpful adding signal to $\mathbf{W}$ becomes.

The scaling exponents we recover in our empirical study with $\tau = 0$ do not match the theory as well as the ones from the experiments in Figure 14 of [PPXP24]. We see two possible reasons for this. First, [PPXP24] do not specify the value of some hyperparameters, namely the input dimension m , the step size γ , the batch size B , and the number of mesh points. We expect that with more extensive hyperparameter tuning, we will be closer to their result. Second, while we only consider C up to 10^9 , [PPXP24] go beyond 10^{10} . We cannot not replicate this due to time limitations for GPU jobs on the server that we use.

4 Discussion

The theory of [PPXP24] provides a detailed description of scaling laws in linear regression under SGD and their dependence on the covariance spectrum of the data distribution and signal. In fact, to our knowledge, it is the first work on the compute-optimal choice of d under SGD dynamics.⁷ The result that the compute-optimal risk decays faster when α and β are larger (and thus the input and signal are for all intents and purposes “lower-dimensional”) is intuitive. The phase transitions between different regimes of power law decay are discussed in other works [LWK⁺24, BAP24a] but at a coarser level and with a less-detailed interpretation. In Section 2, we saw that the phases are differentiated by what combination of four interpretable terms dominates the risk. We have the approximation error ($\mathcal{F}_0$), the error associated with learning the highest-magnitude input and signal directions ($\mathcal{F}_{pp}$), the error associated with learning lower-magnitude input and signal directions ($\mathcal{F}_{ac}$), and the error due to SGD noise ($\mathcal{K}_{pp}$). The compute-optimal choice in each phase arises from trading off the dominant sources of error.

One must proceed with caution when it comes to extrapolating the findings from this work to LLMs, as the assumptions here are far more stringent. For example, it is tempting to look at the results here and postulate that the reason for the large difference between the findings of [KMH⁺20] and Chinchilla scaling [HBM⁺22] is that the respective data distributions have power law spectra with very different exponents. This remains an elusive conjecture that is many steps away from being resolved. The PLRF setting is unable to handle Transformers, but more fundamentally, it cannot handle nonlinearities. A natural next step is the study of scaling laws with single-index teacher models and two-layer neural networks with nonlinear activation function.

Linearity is however not the only limitation of this random matrix theoretic analysis. We highlighted the lack of a rigorous proof that the solution of the deterministic (36) and true Volterra equations (6) are “close”. This implies that there is no rigorous justification that the phases characterized in Figure 4 truly describe the asymptotics of the learning curves and the compute-

⁷[LWK⁺24] do not discuss compute-optimality, [BAP24a] consider learning with gradient flow, and [BAP24a] appeared afterwards.

optimal choice of d . [PPXP24] argue that their empirical results somewhat compensate for this, as they find a good match between their power law fits to the risk curves and their theoretical scaling exponents. We were unable to independently verify this in our simulation study, but in future work, we intend to conduct a more thorough hyperparameter search and incorporate higher orders of magnitude for C . Moreover, Theorem 2.18, the key result that allows $\mathcal{R}$ to asymptotically be expressed as a sum of the dominant components of the forcing and kernel functions, is only proven for $\alpha > 1/4$, but the subsequent derivations in Phases Ib and Ic assume that it holds for all $\alpha \geq 0$.

As for the extension to feature learning, much remains to be explored. In this report, we proposed a simple extension to the PLRF model to incorporate a rank-1 perturbation in the direction of the signal $\mathbf{b}$ into the random feature map $\mathbf{W}$. We motivated this with the recent result by [BES⁺22] showing that a single step of gradient descent on $\mathbf{W}$ is akin to adding such a perturbation under certain assumptions. Though our hope was to find a tractable deterministic equivalent that could easily be related back to the deterministic equivalent in [PPXP24], we are as of yet unable to do so. The key obstacle in this regard is simultaneously handling the power law covariance assumption on the input (which introduces multiple spikes in the spectrum of $\hat{\mathbf{K}}_0$) and the low-rank perturbation (which shifts all of these spikes simultaneously, unlike the canonical spiked model [BBAP05]). We did however find a deterministic equivalent using the method of [HLN07] and ran a simulation to verify the match with the empirical spectrum.

We also wish to highlight the similarity of our feature learning extension to the contemporaneous work [BAP24b]. Our simulation study agrees with their finding that the effect of adding information about the signal into $\mathbf{W}$ is much more pronounced when β is small. At first glance, their model of feature learning, which involves SGD simultaneously on $\mathbf{W}$ and on the parameter vector $\boldsymbol{\theta}$, seems to supersede our simple spiked model. However, their theoretical conclusions are based on non-rigorous DMFT derivations. A fully rigorous characterization of the learning dynamics under the spiked $\mathbf{W}$ model remains valuable and will be the focus of our future work. After having laid the groundwork in Section 3, we see three avenues for approaching this. The first is to plug our $\mathbf{R}$ into the forcing and kernel functions and derive new bounds from scratch. The second is to derive a simpler deterministic equivalent that can be explicitly related to $\mathbf{R}_0$ so that the technical propositions in [PPXP24] can be employed. The third is to forgo deterministic equivalents and return to the original expressions of $\mathcal{F}$ and $\mathcal{K}$ in the proof of Proposition 2.1 as functions of the eigenvalues of $\hat{\mathbf{K}}$.

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Appendix

A SGD Dynamics on Quadratic Functions

This section is dedicated to the proof of Lemma 2.2, which captures the dynamics of a quadratic function of the parameter vector $\boldsymbol{\theta}$ under SGD. We recall the lemma below.

Lemma A.1 (Quadratic Update). *Let $q : \mathbb{R}^d \rightarrow \mathbb{R}$ be a quadratic function. Then, in the setting of Proposition 2.1,*

$$\begin{aligned} \mathbb{E}[q(\boldsymbol{\theta}_{r+1}) - q(\boldsymbol{\theta}_r) | \mathcal{F}_r] &= -\gamma B \langle \nabla q(\boldsymbol{\theta}_r), \mathbf{W}^T \mathbf{D} \mathbf{W} (\boldsymbol{\theta} - \check{\mathbf{b}}) \rangle \\ &\quad + \frac{\gamma^2 B}{2} \langle \nabla^2 q, \mathbf{W}^T \mathbf{D} \mathbf{W} \rangle \| \mathbf{D}^{1/2} (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \|_2^2 \\ &\quad + \gamma^2 B^2 \langle \mathbf{W}^T \mathbf{D} \mathbf{W} (\nabla^2 q) \mathbf{W}^T \mathbf{D} \mathbf{W}, (\boldsymbol{\theta}_r - \check{\mathbf{b}})^{\otimes 2} \rangle. \end{aligned} \quad (97)$$

Proof. We consider a general quadratic function $q : \mathbb{R}^d \rightarrow \mathbb{R}$. Then, by Taylor's Theorem,

$$\mathbb{E}[q(\boldsymbol{\theta}_{r+1}) - q(\boldsymbol{\theta}_r) | \mathcal{F}_r] = \mathbb{E}[\langle \nabla q(\boldsymbol{\theta}_r), \boldsymbol{\theta}_{r+1} - \boldsymbol{\theta}_r \rangle | \mathcal{F}_r] + \frac{1}{2} \mathbb{E}[(\boldsymbol{\theta}_{r+1} - \boldsymbol{\theta}_r)^T (\nabla^2 q) (\boldsymbol{\theta}_{r+1} - \boldsymbol{\theta}_r) | \mathcal{F}_r]. \quad (98)$$

For the first-order term, we plug in the SGD update (3) — using B_r to denote the batch at iteration r — to obtain

$$\begin{aligned} \mathbb{E}[\langle \nabla q(\boldsymbol{\theta}_r), \boldsymbol{\theta}_{r+1} - \boldsymbol{\theta}_r \rangle | \mathcal{F}_r] &= -\gamma \left\langle \nabla q(\boldsymbol{\theta}_r), \mathbb{E} \left[\sum_{j \in B_r} \mathbf{W}^T \mathbf{x}_j (\langle \mathbf{W}^T \mathbf{x}_j, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}_j, \mathbf{b} \rangle) \middle| \mathcal{F}_r \right] \right\rangle \\ &= -\gamma B \langle \nabla q(\boldsymbol{\theta}_r), \mathbf{W}^T \mathbf{D} \mathbf{W} \boldsymbol{\theta}_r - \mathbf{W}^T \mathbf{D} \mathbf{b} \rangle. \end{aligned} \quad (99)$$

For the second order term, we have

$$\begin{aligned} &\frac{1}{2} \mathbb{E}[(\boldsymbol{\theta}_{r+1} - \boldsymbol{\theta}_r)^T (\nabla^2 q) (\boldsymbol{\theta}_{r+1} - \boldsymbol{\theta}_r) | \mathcal{F}_r] \\ &= \frac{1}{2} \mathbb{E}[\langle \nabla^2 q, (\boldsymbol{\theta}_{r+1} - \boldsymbol{\theta}_r)^{\otimes 2} \rangle | \mathcal{F}_r] \\ &= \frac{\gamma^2}{2} \mathbb{E} \left[\left\langle \nabla^2 q, \left(\sum_{j \in B_r} \mathbf{W}^T \mathbf{x}_j (\langle \mathbf{W}^T \mathbf{x}_j, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}_j, \mathbf{b} \rangle) \right)^{\otimes 2} \right\rangle \middle| \mathcal{F}_r \right] \\ &= \frac{\gamma^2}{2} \mathbb{E} \left[\left\langle \nabla^2 q, \sum_{j \in B_r} \left(\mathbf{W}^T \mathbf{x}_j (\langle \mathbf{W}^T \mathbf{x}_j, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}_j, \mathbf{b} \rangle) \right)^{\otimes 2} \right\rangle \middle| \mathcal{F}_r \right] \\ &\quad + \gamma^2 \mathbb{E} \left[\left\langle \nabla^2 q, \sum_{j < k \in B_r} \left(\mathbf{W}^T \mathbf{x}_j (\langle \mathbf{W}^T \mathbf{x}_j, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}_j, \mathbf{b} \rangle) \right) \left(\mathbf{W}^T \mathbf{x}_k (\langle \mathbf{W}^T \mathbf{x}_k, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}_k, \mathbf{b} \rangle) \right)^T \right\rangle \middle| \mathcal{F}_r \right] \\ &= \frac{\gamma^2 B}{2} \mathbb{E}[\langle \nabla^2 q, (\mathbf{W}^T \mathbf{x} [\langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}, \mathbf{b} \rangle])^{\otimes 2} \rangle | \mathcal{F}_r] \\ &\quad + \gamma^2 B(B-1) \mathbb{E}_{\mathbf{x}, \hat{\mathbf{x}} \sim \mathcal{N}_m(\mathbf{0}, \mathbf{D})} [\langle \nabla^2 q, (\mathbf{W}^T \mathbf{x} [\langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}, \mathbf{b} \rangle]) (\mathbf{W}^T \hat{\mathbf{x}} [\langle \mathbf{W}^T \hat{\mathbf{x}}, \boldsymbol{\theta}_r \rangle - \langle \hat{\mathbf{x}}, \mathbf{b} \rangle])^T \rangle | \mathcal{F}_r]. \end{aligned} \quad (100)$$

We decompose $\nabla^2 q = \sum_{j=1}^m \mathbf{v}_j \tilde{\mathbf{v}}_j^T$. Then, we may write

$$\begin{aligned}
& \langle \nabla^2 q, (\mathbf{W}^T \mathbf{x} [\langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}, \mathbf{b} \rangle])^{\otimes 2} \rangle \\
&= (\langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}, \mathbf{b} \rangle)^2 \langle \nabla^2 q, (\mathbf{W}^T \mathbf{x})^{\otimes 2} \rangle \\
&= (\mathbf{x}^T \mathbf{W} \nabla^2 q \mathbf{W}^T \mathbf{x}) [\langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle^2 - 2 \langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle \langle \mathbf{x}, \mathbf{b} \rangle + \langle \mathbf{x}, \mathbf{b} \rangle^2] \\
&= \sum_{j=1}^m \langle \mathbf{x}, \mathbf{W} \mathbf{v}_j \rangle \langle \mathbf{x}, \mathbf{W} \tilde{\mathbf{v}}_j \rangle [\langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle^2 - 2 \langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle \langle \mathbf{x}, \mathbf{b} \rangle + \langle \mathbf{x}, \mathbf{b} \rangle^2].
\end{aligned} \tag{101}$$

To help with taking the expectation of the above, we note that for arbitrary deterministic vectors $\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4 \in \mathbb{R}^m$, we have by Isserlis' Theorem that

$$\begin{aligned}
& \mathbb{E}_{\mathbf{x} \sim \mathcal{N}_m(\mathbf{0}, \mathbf{D})} [\langle \mathbf{x}, \mathbf{u}_1 \rangle \langle \mathbf{x}, \mathbf{u}_2 \rangle \langle \mathbf{x}, \mathbf{u}_3 \rangle \langle \mathbf{x}, \mathbf{u}_4 \rangle] \\
&= \langle \mathbf{D}, \mathbf{u}_1 \mathbf{u}_2^T \rangle \langle \mathbf{D}, \mathbf{u}_3 \mathbf{u}_4^T \rangle + \langle \mathbf{D}, \mathbf{u}_1 \mathbf{u}_3^T \rangle \langle \mathbf{D}, \mathbf{u}_2 \mathbf{u}_4^T \rangle + \langle \mathbf{D}, \mathbf{u}_1 \mathbf{u}_4^T \rangle \langle \mathbf{D}, \mathbf{u}_2 \mathbf{u}_3^T \rangle.
\end{aligned} \tag{102}$$

Taking the expectation of (101) conditional on $\mathcal{F}_r$, we obtain

$$\begin{aligned}
& \mathbb{E} \left[\sum_{j=1}^m \langle \mathbf{W}^T \mathbf{x}, \mathbf{v}_j \rangle \langle \mathbf{W}^T \mathbf{x}, \tilde{\mathbf{v}}_j \rangle \langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle^2 \middle| \mathcal{F}_r \right] \\
&= \sum_{j=1}^m \left[\langle \mathbf{W}^T \mathbf{D} \mathbf{W}, \mathbf{v}_j \tilde{\mathbf{v}}_j^T \rangle \langle \mathbf{W}^T \mathbf{D} \mathbf{W}, \boldsymbol{\theta}_r \boldsymbol{\theta}_r^T \rangle + 2 \langle \mathbf{W}^T \mathbf{D} \mathbf{W}, \mathbf{v}_j \boldsymbol{\theta}_r^T \rangle \langle \mathbf{W}^T \mathbf{D} \mathbf{W}, \tilde{\mathbf{v}}_j \boldsymbol{\theta}_r^T \rangle \right]
\end{aligned} \tag{103}$$

for the first term,

$$\begin{aligned}
& \mathbb{E} \left[-2 \sum_{j=1}^m \langle \mathbf{x}, \mathbf{W} \mathbf{v}_j \rangle \langle \mathbf{x}, \mathbf{W} \tilde{\mathbf{v}}_j \rangle \langle \mathbf{x}, \mathbf{W} \boldsymbol{\theta}_r \rangle \langle \mathbf{x}, \mathbf{b} \rangle \middle| \mathcal{F}_r \right] \\
&= -2 \sum_{j=1}^m \left[\langle \mathbf{W}^T \mathbf{D} \mathbf{W}, \mathbf{v}_j \tilde{\mathbf{v}}_j^T \rangle \langle \mathbf{D}, \mathbf{W} \boldsymbol{\theta}_r \otimes \mathbf{b} \rangle + \langle \mathbf{D}, \mathbf{W} \mathbf{v}_j \otimes \mathbf{W} \boldsymbol{\theta}_r \rangle \langle \mathbf{D}, \mathbf{W} \tilde{\mathbf{v}}_j \otimes \mathbf{b} \rangle \right. \\
&\quad \left. + \langle \mathbf{D}, \mathbf{W} \mathbf{v}_j \otimes \mathbf{b} \rangle \langle \mathbf{D}, \mathbf{W} \tilde{\mathbf{v}}_j \otimes \mathbf{W} \boldsymbol{\theta}_r \rangle \right]
\end{aligned} \tag{104}$$

for the second term, and

$$\begin{aligned}
& \mathbb{E} \left[\sum_{j=1}^m \langle \mathbf{x}, \mathbf{W} \mathbf{v}_j \rangle \langle \mathbf{x}, \mathbf{W} \tilde{\mathbf{v}}_j \rangle \langle \mathbf{x}, \mathbf{b} \rangle^2 \middle| \mathcal{F}_r \right] \\
&= \sum_{j=1}^m \left[\langle \mathbf{W}^T \mathbf{D} \mathbf{W}, \mathbf{v}_j \tilde{\mathbf{v}}_j^T \rangle \langle \mathbf{D}, \mathbf{b} \mathbf{b}^T \rangle + 2 \langle \mathbf{D}, \mathbf{W} \mathbf{v}_j \otimes \mathbf{b} \rangle \langle \mathbf{D}, \mathbf{W} \tilde{\mathbf{v}}_j \otimes \mathbf{b} \rangle \right]
\end{aligned} \tag{105}$$

for the third term.

$$\begin{aligned}
& \mathbb{E} \left[\sum_{j=1}^m \langle \mathbf{x}, \mathbf{W} \mathbf{v}_j \rangle \langle \mathbf{x}, \mathbf{W} \tilde{\mathbf{v}}_j \rangle [\langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle^2 - 2 \langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle \langle \mathbf{x}, \mathbf{b} \rangle + \langle \mathbf{x}, \mathbf{b} \rangle^2] \right] \\
&= \langle \nabla^2 q, \mathbf{W}^T \mathbf{D} \mathbf{W} \rangle \| \mathbf{D}^{1/2} (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \|_2^2 \\
&\quad + 2 \sum_{j=1}^m \left[\langle \mathbf{D}, \mathbf{W} \mathbf{v}_j \otimes \mathbf{W} \boldsymbol{\theta}_r \rangle \langle \mathbf{D}, \mathbf{W} \tilde{\mathbf{v}}_j \otimes \mathbf{W} \boldsymbol{\theta}_r \rangle - \langle \mathbf{D}, \mathbf{W} \mathbf{v}_j \otimes \mathbf{W} \boldsymbol{\theta}_r \rangle \langle \mathbf{D}, \mathbf{W} \tilde{\mathbf{v}}_j \otimes \mathbf{b} \rangle \right. \\
&\quad \left. + \langle \mathbf{D}, \mathbf{W} \mathbf{v}_j \otimes \mathbf{b} \rangle \langle \mathbf{D}, \mathbf{W} \tilde{\mathbf{v}}_j \otimes \mathbf{b} \rangle - \langle \mathbf{D}, \mathbf{W} \tilde{\mathbf{v}}_j \otimes \mathbf{W} \boldsymbol{\theta}_r \rangle \langle \mathbf{D}, \mathbf{W} \mathbf{v}_j \otimes \mathbf{b} \rangle \right] \\
&= \langle \nabla^2 q, \mathbf{W}^T \mathbf{D} \mathbf{W} \rangle \| \mathbf{D}^{1/2} (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \|_2^2 \\
&\quad + 2 \sum_{j=1}^m \langle \mathbf{D}, \mathbf{W} \mathbf{v}_j \otimes (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \rangle \langle \mathbf{D}, \mathbf{W} \tilde{\mathbf{v}}_j \otimes (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \rangle.
\end{aligned} \tag{106}$$

The last term is conveniently simplified as

$$\begin{aligned}
& \sum_{j=1}^m \langle \mathbf{D}, \mathbf{W} \mathbf{v}_j \otimes (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \rangle \langle \mathbf{D}, \mathbf{W} \tilde{\mathbf{v}}_j \otimes (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \rangle \\
&= \sum_{j=1}^m \langle \mathbf{W}^T \mathbf{D}, \mathbf{v}_j \otimes (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \rangle \langle \mathbf{W}^T \mathbf{D}, \tilde{\mathbf{v}}_j \otimes (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \rangle \\
&= \sum_{j=1}^m \sum_{i,j,k,l} (W^T D)_{ni} v_{jn} (W \boldsymbol{\theta}_r - \mathbf{b})_i (W^T D)_{kl} \tilde{v}_{jk} (W \boldsymbol{\theta}_r - \mathbf{b})_l \\
&= 2 \langle \mathbf{D} \mathbf{W} (\nabla^2 q) \mathbf{W}^T \mathbf{D}, (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})^{\otimes 2} \rangle.
\end{aligned} \tag{107}$$

Now, returning to the second term in the last equation of (100), we have (using properties of the Frobenius inner product)

$$\begin{aligned}
& \mathbb{E} \left[\langle \nabla^2 q, (\mathbf{W}^T \mathbf{x} [\langle \mathbf{W}^T \mathbf{x}, \boldsymbol{\theta}_r \rangle - \langle \mathbf{x}, \mathbf{b} \rangle]) (\mathbf{W}^T \hat{\mathbf{x}} [\langle \mathbf{W}^T \hat{\mathbf{x}}, \boldsymbol{\theta}_r \rangle - \langle \hat{\mathbf{x}}, \mathbf{b} \rangle])^T \rangle \middle| \mathcal{F}_r \right] \\
&= \mathbb{E} [\langle \nabla^2 q, \mathbf{W}^T \mathbf{x} \mathbf{x}^T (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})^T \hat{\mathbf{x}} \hat{\mathbf{x}}^T \mathbf{W} \rangle | \mathcal{F}_r] \\
&= \mathbb{E} [(\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})^T \mathbf{x} \mathbf{x}^T \mathbf{W} \nabla^2 q \mathbf{W}^T \hat{\mathbf{x}} \hat{\mathbf{x}}^T (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) | \mathcal{F}_r] \\
&= (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})^T \mathbf{D} \mathbf{W} \nabla^2 q \mathbf{W}^T \mathbf{D} (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \\
&= \langle \mathbf{D} \mathbf{W} (\nabla^2 q) \mathbf{W}^T \mathbf{D}, (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})^{\otimes 2} \rangle.
\end{aligned} \tag{108}$$

Finally, plugging everything back into (100), we have

$$\begin{aligned}
\frac{1}{2} \mathbb{E} [\langle \nabla^2 q, (\boldsymbol{\theta}_{r+1} - \boldsymbol{\theta}_r)^{\otimes 2} \rangle | \mathcal{F}_r] &= \frac{\gamma^2 B}{2} \langle \nabla^2 q, \mathbf{W}^T \mathbf{D} \mathbf{W} \rangle \| \mathbf{D}^{1/2} (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \|_2^2 \\
&\quad + \gamma^2 B^2 \langle \mathbf{D} \mathbf{W} (\nabla^2 q) \mathbf{W}^T \mathbf{D}, (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})^{\otimes 2} \rangle.
\end{aligned} \tag{109}$$

Hence, combining (99) and (109), we obtain

$$\begin{aligned}
\mathbb{E} [q(\boldsymbol{\theta}_{r+1}) - q(\boldsymbol{\theta}_r) | \mathcal{F}_r] &= -\gamma B \langle \nabla q(\boldsymbol{\theta}_r), \mathbf{W}^T \mathbf{D} \mathbf{W} \boldsymbol{\theta}_r - \mathbf{W}^T \mathbf{D} \mathbf{b} \rangle \\
&\quad + \frac{\gamma^2 B}{2} \langle \nabla^2 q, \mathbf{W}^T \mathbf{D} \mathbf{W} \rangle \| \mathbf{D}^{1/2} (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \|_2^2 \\
&\quad + \gamma^2 B^2 \langle \mathbf{D} \mathbf{W} (\nabla^2 q) \mathbf{W}^T \mathbf{D}, (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b})^{\otimes 2} \rangle.
\end{aligned} \tag{110}$$

Leveraging our decomposition of $\mathbf{b}$ in (11), we may re-write the above as

$$\begin{aligned}\mathbb{E}[q(\boldsymbol{\theta}_{r+1}) - q(\boldsymbol{\theta}_r) | \mathcal{F}_r] &= -\gamma B \langle \nabla q(\boldsymbol{\theta}_r), \mathbf{W}^T \mathbf{D} \mathbf{W} (\boldsymbol{\theta} - \check{\mathbf{b}}) \rangle \\ &\quad + \frac{\gamma^2 B}{2} \langle \nabla^2 q, \mathbf{W}^T \mathbf{D} \mathbf{W} \rangle \| \mathbf{D}^{1/2} (\mathbf{W} \boldsymbol{\theta}_r - \mathbf{b}) \|_2^2 \\ &\quad + \gamma^2 B^2 \langle \mathbf{W}^T \mathbf{D} \mathbf{W} (\nabla^2 q) \mathbf{W}^T \mathbf{D} \mathbf{W}, (\boldsymbol{\theta}_r - \check{\mathbf{b}})^{\otimes 2} \rangle.\end{aligned}\tag{111}$$

□

B Asymptotics of $\mathcal{F}_0$

We re-state Proposition 2.10 and detail its proof here.

Proposition B.1 (Asymptotics of $\mathcal{F}_0$, [PPXP24] Proposition 10.3). *Assume $2\alpha + 2\beta > 1$. Then,*

$$\mathcal{F}_0(r) \asymp \begin{cases} \frac{1}{\kappa} d^{-2\alpha} \sum_{j=1}^m j^{-2\beta} & \text{if } 2\beta > 1, \\ d^{1-2\alpha-2\beta} \int_0^{m/d} \frac{u^{-2\beta}}{\kappa+u^{2\alpha}} du & \text{if } 2\beta < 1. \end{cases} \asymp d^{-2\alpha+(1-2\beta)_+}.\tag{112}$$

Proof. We follow the proof in [PPXP24] and fill in some small details. We split the proof into two cases based on the value of β .

Case $2\beta > 1$: Let $\tilde{C} = \sum_{j=1}^m j^{-2\beta}$. Consider the relative error

$$\begin{aligned}\mathcal{E}_1(r) &:= \frac{\left| \sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{j^{-2\alpha}\kappa d^{2\alpha}+1} - \frac{d^{-2\alpha}}{\kappa} \sum_{j=1}^m j^{-2\beta} \right|}{\frac{d^{-2\alpha}\tilde{C}}{\kappa}} \\ &= \frac{1}{\tilde{C}} \left| \sum_{j=1}^m \frac{\kappa d^{2\alpha} j^{-2\alpha-2\beta}}{j^{-2\alpha}\kappa d^{2\alpha}+1} - \sum_{j=1}^m \frac{j^{-2\beta} + j^{-2\alpha-2\beta} \kappa d^{2\alpha}}{j^{-2\alpha}\kappa d^{2\alpha}+1} \right| \\ &= \frac{1}{\tilde{C}} \sum_{j=1}^m \frac{j^{-2\beta}}{j^{-2\alpha}\kappa d^{2\alpha}+1}.\end{aligned}\tag{113}$$

Since $2\beta > 1$, for any $\varepsilon > 0$ there exists an index j_0 such that

$$\frac{1}{\tilde{C}} \sum_{j=j_0}^m \frac{j^{-2\beta}}{j^{-2\alpha}\kappa d^{2\alpha}+1} \leq \frac{1}{\tilde{C}} \sum_{j=j_0}^m j^{-2\beta} < \varepsilon.\tag{114}$$

Meanwhile, the first j_0 terms in the sum,

$$\sum_{j=1}^{j_0} \frac{j^{-2\beta}}{j^{-2\alpha}\kappa d^{2\alpha}+1} \leq \sum_{j=1}^{j_0} \frac{j^{-2\beta}}{j_0^{-2\alpha}\kappa d^{2\alpha}+1} \leq \frac{j_0}{j_0^{-2\alpha}\kappa d^{2\alpha}+1},\tag{115}$$

vanish as $d \rightarrow \infty$.

Hence,

$$\left| \sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{j^{-2\alpha}\kappa d^{2\alpha}+1} - \frac{d^{-2\alpha}}{\kappa} \sum_{j=1}^m j^{-2\beta} \right| = o(d^{-2\alpha}).\tag{116}$$

Case $2\beta < 1$: We consider the two errors

$$\mathcal{E}_{21}(r) := \frac{\left| \sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{j^{-2\alpha}\kappa d^{2\alpha}+1} - d^{1-2\alpha-2\beta} \int_{1/d}^{m/d} \frac{u^{-2\beta}}{\kappa+u^{2\alpha}} du \right|}{d^{1-2\alpha-2\beta} \int_0^{m/d} \frac{u^{-2\beta}}{\kappa+u^{2\alpha}} du} \quad (117)$$

and

$$\mathcal{E}_{22}(r) := \frac{\left| d^{1-2\alpha-2\beta} \int_{1/d}^{m/d} \frac{u^{-2\beta}}{\kappa+u^{2\alpha}} du - d^{1-2\alpha-2\beta} \int_0^{m/d} \frac{u^{-2\beta}}{\kappa+u^{2\alpha}} du \right|}{d^{1-2\alpha-2\beta} \int_0^{m/d} \frac{u^{-2\beta}}{\kappa+u^{2\alpha}} du}. \quad (118)$$

By the Monotone Convergence Theorem, it is immediate that $\mathcal{E}_{22}(r) \rightarrow 0$ as $d \rightarrow \infty$. Now, we rewrite

$$\sum_{j=1}^m \frac{j^{-2\alpha-2\beta}}{j^{-2\alpha}\kappa d^{2\alpha}+1} = d^{1-2\alpha-2\beta} \frac{1}{d} \sum_{j=1}^m \frac{(j/d)^{-2\alpha-2\beta}}{(j/d)^{-2\alpha}\kappa+1} \quad (119)$$

and make use of the integral approximation in the result below.

Lemma B.2 (Trapezoidal Rule, [PPXP24] Proposition 10.1). *Assume f is continuous, $a < b \in \mathbb{R}$, $n \in \mathbb{N}$, and define*

$$T_n(f) := \frac{b-a}{2n} (f(x_0) + 2f(x_1) + \cdots + 2f(x_{n-1}) + f(x_n)), \quad (120)$$

where $x_i = a + i(b-a)/n$, $0 \leq i \leq n$. If f has an integrable first derivative, then

$$\left| T_n(f) - \int_a^b f(t) dt \right| \leq \frac{b-a}{n} \int_a^b |f'(t)| dt. \quad (121)$$

We take $a = 1/d$, $b = m/d$, $n = m-1$, $x_j = 1/d + j/d$ for $0 \leq j \leq n$, and $f(x) = \frac{x^{-2\alpha-2\beta}}{x^{-2\alpha}\kappa+1}$. We have

$$T_n(f) = \frac{1}{d} \sum_{j=1}^m \frac{(1/d)^{-2\alpha-2\beta}}{(j/d)^{-2\alpha}\kappa+1} - \frac{1}{2d} \left(\frac{(1/d)^{-2\alpha-2\beta}}{(1/d)^{-2\alpha}\kappa+1} + \frac{(m/d)^{-2\alpha-2\beta}}{(m/d)^{-2\alpha}\kappa+1} \right). \quad (122)$$

By the lemma,

$$\mathcal{E}_{21}(r) \leq \frac{\frac{1}{d} \int_{1/d}^{m/d} |f'(x)| dx + \frac{1}{2d} \left(\frac{(\frac{1}{d})^{-2\alpha-2\beta}}{(\frac{1}{d})^{-2\alpha}\kappa+1} + \frac{(\frac{m}{d})^{-2\alpha-2\beta}}{(\frac{m}{d})^{-2\alpha}\kappa+1} \right)}{\int_0^{m/d} \frac{u^{-2\beta}}{\kappa+u^{2\alpha}} du}. \quad (123)$$

Note that the denominator is of constant order due to our assumption on the ratio m/d . The second term thus vanishes as $d \rightarrow \infty$ since $2\beta < 1$. For the first term, notice that

$$\begin{aligned} f'(x) &= \frac{(-2\alpha-2\beta)x^{-2\alpha-2\beta-1}(x^{-2\alpha}\kappa+1) + 2\alpha x^{-2\alpha-1}\kappa x^{-2\alpha-2\beta}}{(x^{-2\alpha}\kappa+1)^2} \\ &= -\frac{(2\alpha+2\beta)x^{-2\alpha-2\beta-1} + 2\beta\kappa x^{-4\alpha-2\beta-1}}{(x^{-2\alpha}\kappa+1)^2}, \end{aligned} \quad (124)$$

which is negative for all $x \in [a, b]$. Hence, $|f'(x)| = -f'(x)$ and

$$\int_{1/d}^{m/d} |f'(x)| dx = f(1/d) - f(m/d) = \frac{(\frac{1}{d})^{-2\alpha-2\beta}}{(\frac{1}{d})^{-2\alpha}\kappa+1} - \frac{(\frac{m}{d})^{-2\alpha-2\beta}}{(\frac{m}{d})^{-2\alpha}\kappa+1} = O(d^{2\beta}). \quad (125)$$

Therefore, since $2\beta < 1$, the first term vanishes as $d \rightarrow \infty$. $\square$

C Expanding the Spiked W

Under the setting of Section 3, recall the expansion

$$\begin{aligned}\hat{\mathbf{K}} &= \mathbf{D}^{1/2} \mathbf{W} \mathbf{W}^T \mathbf{D}^{1/2} = \mathbf{D}^{1/2} (\tau \mathbf{b} \mathbf{1}^T + \mathbf{Z}) (\tau \mathbf{b} \mathbf{1}^T + \mathbf{Z})^T \mathbf{D}^{1/2} \\ &= \tau^2 d \mathbf{a} \mathbf{a}^T + \tau \mathbf{a} \mathbf{1}^T \mathbf{Z}^T \mathbf{D}^{1/2} + \tau \mathbf{D}^{1/2} \mathbf{Z} \mathbf{1} \mathbf{a}^T + \mathbf{D}^{1/2} \mathbf{Z} \mathbf{Z}^T \mathbf{D}^{1/2}.\end{aligned}\quad (126)$$

We analyze the magnitude of each term.

Importantly, we have that $\|\mathbf{Z}\| = \Theta(1)$ with high probability (i.e., with probability tending to 1 as $m, d \rightarrow \infty$). This follows from standard lower and upper bounds on the singular values of a random matrix with i.i.d. Gaussian entries (e.g., [Ver18] Theorem 2.6.1). Hence, $\|\hat{\mathbf{K}}_0\| = \Theta(1)$ w.h.p. Moreover,

$$\|\mathbf{D}^{1/2} \mathbf{Z} \mathbf{1} \mathbf{a}^T\| \lesssim \|\mathbf{Z}\| \|\mathbf{1} \mathbf{a}^T\| = \|\mathbf{Z}\| = \Theta(1) \quad (127)$$

with high probability. For the least term, it is easily seen that $\|\mathbf{a} \mathbf{a}^T\| = \|\mathbf{a}\|_2^2$.

This motivates taking $\tau = \Theta(1/\sqrt{d})$. Then $\hat{\mathbf{K}}_0$ and the rank-one update $\tau^2 d \mathbf{a} \mathbf{a}^T$ are of constant order, while the cross terms vanish with d . Hence, it is sufficient to consider a deterministic equivalent for $\mathbf{K} = \hat{\mathbf{K}}_0 + \tau^2 d \mathbf{a} \mathbf{a}^T$. Indeed, by the resolvent identity (e.g., [CL22] Lemma 2.1), for all $z \in \mathbb{C} \setminus (\text{spec}(\mathbf{K}) \cup \text{spec}(\hat{\mathbf{K}}))$,

$$(\mathbf{K} - z)^{-1} - (\hat{\mathbf{K}} - z)^{-1} = (\mathbf{K} - z)^{-1} (\hat{\mathbf{K}} - \mathbf{K}) (\hat{\mathbf{K}} - z)^{-1}. \quad (128)$$

Hence,

$$\|(\mathbf{K} - z)^{-1} - (\hat{\mathbf{K}} - z)^{-1}\| \leq \left(\frac{1}{\text{dist}(z, \text{spec}(\mathbf{K}))} \right) \left(\frac{1}{\text{dist}(z, \text{spec}(\hat{\mathbf{K}}))} \right) (2\tau \|\mathbf{D}^{1/2} \mathbf{Z} \mathbf{1} \mathbf{a}^T\|), \quad (129)$$

which vanishes as $m, d \rightarrow \infty$. It is then immediate by Hölder's Inequality that $\mathbf{K}$ and $\hat{\mathbf{K}}$ are equivalent.

Now, by Sherman-Morrison,

$$(\mathbf{K} - z)^{-1} = (\hat{\mathbf{K}}_0 + \tau^2 d \mathbf{a} \mathbf{a}^T - z)^{-1} = (\hat{\mathbf{K}}_0 - z)^{-1} - \frac{\tau^2 d (\hat{\mathbf{K}}_0 - z)^{-1} \mathbf{a} \mathbf{a}^T (\hat{\mathbf{K}}_0 - z)^{-1}}{1 + \tau^2 d \mathbf{a}^T (\hat{\mathbf{K}}_0 - z)^{-1} \mathbf{a}}. \quad (130)$$

Taking the trace, we have

$$\frac{1}{m} \text{tr}(\mathbf{K} - z)^{-1} = \frac{1}{m} \text{tr}(\hat{\mathbf{K}}_0 - z)^{-1} - \frac{1}{m} \frac{\tau^2 d \mathbf{a}^T (\hat{\mathbf{K}}_0 - z)^{-2} \mathbf{a}}{1 + \tau^2 d \mathbf{a}^T (\hat{\mathbf{K}}_0 - z)^{-1} \mathbf{a}}. \quad (131)$$

But the last term vanishes as $m \rightarrow \infty$. Hence, Both $\hat{\mathbf{K}}_0$ and $\mathbf{K}$ have the *same* deterministic equivalent. This is in fact not surprising, as this result is well known from the study of spiked models and the BBP phase transition [BBAP05]. The spectral bulk of $\hat{\mathbf{K}}_0$ and $\mathbf{K}$ are identical. The only difference is the emergence of a spike for sufficiently large τ .

D Numerical Solution for Spiked Deterministic Equivalent

Consider the system of equations.

$$f_1(s, t) := s + \frac{d}{z}u + \frac{\frac{\tau^2}{z^2}du^2\mathbf{a}^T\mathbf{S}\mathbf{a}}{1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a}} = 0 \quad (132)$$

$$f_2(s, t) := t + \text{tr } \mathbf{D} \left(\frac{1}{z}\mathbf{S} + \frac{\frac{\tau^2}{z^2}du\mathbf{S}\mathbf{a}\mathbf{a}^T\mathbf{S}}{1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a}} \right) = t + \frac{1}{z} \text{tr } \mathbf{D}\mathbf{S} + \frac{\frac{\tau^2}{z^2}du\mathbf{a}^T\mathbf{D}\mathbf{S}^2\mathbf{a}}{1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a}} = 0. \quad (133)$$

We solve this system via Newton's method. Note that

$$\mathbf{a}^T(\mathbf{I}_v + \frac{s}{d}\mathbf{D})^{-1}\mathbf{a} = \sum_{j=1}^v \frac{a_j^2}{1 + \frac{s}{d}j^{-2\alpha}},$$

and therefore

$$\frac{d}{ds}[\mathbf{a}^T\mathbf{S}\mathbf{a}] = -\frac{1}{d} \sum_{j=1}^m \frac{a_j^2}{(1 + \frac{s}{d}j^{-2\alpha})^2} j^{-2\alpha} = -\frac{1}{d} \mathbf{a}^T \mathbf{D} \mathbf{S}^2 \mathbf{a}. \quad (134)$$

We proceed with

$$\begin{aligned} \frac{df_1}{ds} &= 1 - \frac{\frac{\tau^2}{z^2}u^2(1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a})\mathbf{a}^T\mathbf{D}\mathbf{S}^2\mathbf{a}}{(1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a})^2} - \frac{\frac{\tau^2}{z^2}du^2\mathbf{a}^T\mathbf{S}\mathbf{a}(\frac{\tau^2}{z}u\mathbf{a}^T\mathbf{D}\mathbf{S}^2\mathbf{a})}{(1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a})^2} \\ &= 1 - \frac{\frac{\tau^2}{z^2}u^2\mathbf{a}^T\mathbf{D}\mathbf{S}^2\mathbf{a}}{(1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a})^2}. \end{aligned}$$

$$\mathbf{a}^T \mathbf{D} \mathbf{S}^2 \mathbf{a} = \sum_{j=1}^m \frac{a_j^2}{(1 + \frac{s}{d}j^{-2\alpha})^2} j^{-2\alpha} \quad (135)$$

$$\Rightarrow \frac{d}{ds}[\mathbf{a}^T \mathbf{D} \mathbf{S}^2 \mathbf{a}] = -\frac{2}{d} \sum_{j=1}^m \frac{a_j^2}{(1 + \frac{s}{d}j^{-2\alpha})^3} j^{-4\alpha} = -\frac{2}{d} \mathbf{a}^T \mathbf{D}^2 \mathbf{S}^3 \mathbf{a}. \quad (136)$$

$$\text{tr } \mathbf{D}\mathbf{S} = \sum_{j=1}^m \frac{j^{-2\alpha}}{(1 + \frac{s}{d}j^{-2\alpha})} \quad (137)$$

$$\Rightarrow \frac{d}{ds}[\text{tr } \mathbf{D}\mathbf{S}] = -\frac{1}{d} \sum_{j=1}^m \frac{j^{-4\alpha}}{(1 + \frac{s}{d}j^{-2\alpha})^2} = -\frac{1}{d} \text{tr } \mathbf{D}^2 \mathbf{S}^2. \quad (138)$$

$$\frac{df_2}{ds} = -\frac{1}{zd} \text{tr } \mathbf{D}^2 \mathbf{S}^2 - \frac{2\frac{\tau^2}{z^2}u\mathbf{a}^T\mathbf{D}^2\mathbf{S}^3\mathbf{a}(1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a}) - \frac{\tau^2}{z^2}du\mathbf{a}^T\mathbf{D}\mathbf{S}^2\mathbf{a}(\frac{\tau^2}{z}u\mathbf{a}^T\mathbf{D}\mathbf{S}^2\mathbf{a})}{(1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a})^2} \quad (139)$$

$$\frac{du}{dt} = -\frac{1}{d} \frac{1}{(1 + \frac{t}{d})^2} = -\frac{u^2}{d}. \quad (140)$$

$$\begin{aligned} \frac{df_1}{dt} &= -\frac{u^2}{z} - \frac{2\frac{\tau^2}{z^2}u^3\mathbf{a}^T\mathbf{S}\mathbf{a}(1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a})}{(1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a})^2} - \frac{\frac{\tau^2}{z^2}du^2\mathbf{a}^T\mathbf{S}\mathbf{a}(\frac{\tau^2}{z}u^2\mathbf{a}^T\mathbf{S}\mathbf{a})}{(1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a})^2} \\ &= -\frac{u^2}{z} + \frac{\frac{\tau^4}{z^3}du^4(\mathbf{a}^T\mathbf{S}\mathbf{a})^2 - 2\frac{\tau^2}{z^2}u^3\mathbf{a}^T\mathbf{S}\mathbf{a}}{(1 - \frac{\tau^2}{z}du\mathbf{a}^T\mathbf{S}\mathbf{a})^2}. \end{aligned} \quad (141)$$

$$\begin{aligned}
\frac{df_2}{dt} &= 1 - \frac{\frac{\tau^2}{z^2} u^2 \mathbf{a}^T \mathbf{D} \mathbf{S}^2 \mathbf{a} (1 - \frac{\tau^2}{z} du \mathbf{a}^T \mathbf{S} \mathbf{a})}{(1 - \frac{\tau^2}{z} du \mathbf{a}^T \mathbf{S} \mathbf{a})} - \frac{\frac{\tau^2}{z^2} du \mathbf{a}^T \mathbf{D} \mathbf{S}^2 \mathbf{a} (\frac{\tau^2}{z} u^2 \mathbf{a}^T \mathbf{S} \mathbf{a})}{(1 - \frac{\tau^2}{z} du \mathbf{a}^T \mathbf{S} \mathbf{a})^2} \\
&= 1 - \frac{\frac{\tau^2}{z^2} u^2 \mathbf{a}^T \mathbf{D} \mathbf{S}^2 \mathbf{a}}{(1 - \frac{\tau^2}{z} du \mathbf{a}^T \mathbf{S} \mathbf{a})^2}.
\end{aligned} \tag{142}$$